QUALIFICATION STANDARD AND PLAN

USQ Electrical Maintainer VIC

Marine Technician - Electrical
(MAR TECH - Elec) 00379-03

NQUAL – AIMM

MITE ID –
Residential - 123038
OJPR - 123041

RECORD OF AMENDMENTS

1. The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

2. Qualification Standard and Plan (QSP) versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents. Versions with a number followed by a bracketed number indicate a stray from this nomenclature. The number in brackets indicates the corrected version number.

Version Number	Amendment Type	Approval Date	Authority
1.0			
3.0	3-CH Conversion	21 Aug 2022	A/CNPTG, Cdr Stefanson

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ACRONYMS AND ABBREVIATIONS

A	
AC	Alternating Current
ACU	Automatic Control Unit
ARR	All Round Reading
AT	Agitation Test
ATT	Actual Training Time
AVFR	Automatic Voltage Frequency Regulator
AVR	Automatic Voltage Regulator
B	
BAT	Base Administrative Time
BR	Book of Reference
C	
CAF	Canadian Armed Forces
CCA	Curriculum Control Authority
CFAO	Canadian Forces Administrative Order
CFITES	Canadian Forces Individual Training and Education System
CM	Corrective Maintenance
Cmdt	Commandant
CMS	Chief of Maritime Staff
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
COTM	Commanding Officer Temporary Memorandum
CSNI	Consolidated Secret Network Infrastructure
D	
DAOD	Defence Administrative Orders and Directives
DC	Direct Current
DEMO	Demonstration
DIN	Defense Intranet Network
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DPGR	Director Personnel Generation Requirements
DWAN	Defence Wide Area Network
D4	Quarterly Discharge
E	
EC	Enabling Check
EO	Enabling Objective
EOTI	Engineering Officer Temporary Instruction
EOTM	Engineering Officer Temporary Memorandum
G	
GPNet	General Purpose Network
H	
HOD	Head of Department
HP	High Pressure
Hz	Hertz

I	
IAW	In Accordance With
ID	Identification
IMS	Information Management Standard
IS	Information Systems
IT	Information Technology
K	
kW	Kilowatt
L	
LCP	Local Control Panel
LG	Lesson Guidance
LO	Learning Objective
LP	Low Power
LS	Lesson Specification
M	
MARCORD	Maritime Command Order
MCC	Machinery Control Console
MITE	Military Individual Training and Education
MOSID	Military Occupational Structure Identification Code
MPV	Minimum Permissible Voltage
MRDB	Maintenance Record Database
MTPU	Motor Thermal Protection Unit
N	
NCM	Non-Commissioned Member
NQual	National Qualification
NFS	Naval Fleet School
NAVORD	Naval Order
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order
NTDC(A)	Naval Training Development Centre (Atlantic)
NTI	Naval Training Instruction
O	
OJPR	On-the-Job Performance Requirement
OJT	On-the-Job Training
OMC	One Man Control
OS	Occupation Specification
OSS	Occupation Specialty Specification
OST	Out Serviced Training
P	
PC	Performance Check
PLAR	Prior Learning Assessment and Recognition
PM	Preventive Maintenance
PMG	Permanent Magnet Generator
PO	Performance Objective
PP&S	Paper, Pens and Stationary

PPE	Personal Protective Equipment
PRR	Personal Record Resume
PT	Physical Training
Q	
QL	Qualification Level
QMS/Prog Rep	Qualification Management System/ Progress Report
QRS	Qualification Record Sheet
QSP	Qualification Standard and Plan
R	
R&Q	Rations & Quarters
RCMS	Reliability Centered Maintenance System
RCN	Royal Canadian Navy
S	
SAT	Satisfactory
SCC	Ship Control Console
SO	Standing Order
SOP	Standard Operating Procedure
SSMG	Slow Speed Motor Generator
T	
TAT	Training Administrative Time
TA	Training Authority
TCC	Tightness Connection Check
TD	Temporary Duty
TDP	Technical Data Package
TE	Training Establishment
TEL	Technology Enabled Learning
TO	Training Officer
TP	Teaching Point
TPO	Training Petty Officer
TRB	Training Review Board
TRU	Transformer Rectifier Unit
TS	Training Site
TTT	Training Travel Time
TU	Top Up
U	
UER	Unit Employment Record
UNSAT	Unsatisfactory
V	
V	Volt
VCSSO	Victoria Class Submarine Standing Order
VCSTI	Victoria Class Submarine Technical Instruction
Vol	Volume
VPDC	Variable Pressure Direct Current

FOREWORD

1. This training is based on the specification found at Director Personnel Generation Requirements (DPGR) website: **http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page**. and:
   a. A-PD-055-002/PP-002; and
   b. A-PD-055-003/PQ-001.

2. A trainee will be granted the National Qualification (NQual) AIMM when the standards listed in all POs have been met.

3. A trainee will be granted the following Military Individual Training and Education (MITE) identifications (IDs) when standards listed in all POs have been met:

   a. Residential: POs 001 – 009 - 123038
   b. OJPR - 123041

4. The designated TEs for this training include NFS(A) and NFS(P) within NPTG.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this QSP is to establish the requirements and training for the USQ Electrical Maintainer VIC (AIMM). This QSP is designed to assist NTDC(A) in developing, NFSs in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) Naval Personnel and Training Order (NPTORD) 4500-22 Annex B Table 3.1. Partial days are rounded up. For this course, a period of instruction is 50 minutes in duration and each day has 8 instructional periods. The course duration calculation is based on maximum load.

Course Duration Calculation		
Allocation of Actual Training Time (ATT)	Periods	Days
Lesson Guidance (LG)	118	14.75
Enabling Checks (ECs)	25	3.125
Performance Checks (PCs)	6	.75
ATT SUB TOTAL:	**149**	**18.625**
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	2	.25
Training Administrative Time (TAT)	2	.25
Training Travel Time (TTT)	0	0
Physical Fitness (PT)	26	3.25
BAT, TAT, TTT, PT SUB TOTAL:	**30**	**3.75**
***Partial days are rounded up** **TOTAL:**	**179**	**22.375**

REQUIRED COURSE STAFFING

3. The following personnel, including guest lecturers, are required to conduct this training:

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Course Officer	Lt(N)	1	00345		
Senior Instructor	PO1	1	AILS, AIND, AIMM		
Instructor	MS/PO2	2	AILS, AIND, AIMM		
FTA/STO	Lt(N)/LCdr	1	N/A		

COURSE CAPACITY

4. Planning for the conduct of training is based on availability of resources; course capacities are as follows:

5. Maximum – 6;

6. Minimum – 2; and

7. Optimum – 4.

PREREQUISITES

8. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

9. USQ S1 Electrical (ALUT); and

10. Basic Submarine Qualification (AILS).

PRIOR LEARNING ASSESSMENT AND RECOGNITION

11. Prior Learning Assessment and Recognition (PLARs) shall be conducted IAW Naval Order (NAVORD) 4500-2, Defence Administrative Orders and Directives (DAOD) 5031-1, and Canadian Forces Individual Training and Education System (CFITES) Volume (Vol) 12.

SPECIAL INSTRUCTIONS

12. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – MAINTAIN MAIN BATTERY & ASSOCIATED SYSTEMS

13. **Performance**: Maintain Main Battery & Associated Systems.

14. **Conditions**:

    a. Given:

        (1) References;
        (2) Main battery & associated systems;
        (3) Electrical tools;
        (4) Cut out devices;
        (5) Battery log;
        (6) Safety equipment;
        (7) PPE; and
        (8) Test equipment.

    b. Denied: Nil; and

    c. Environment:

        (1) Operational submarine; and
        (2) Alongside or at sea.

15. **Standard**: The Maintainer shall Maintain Main Battery & Associated Systems by:

    a. Performing battery PM, IAW all refs and safety precautions, by:

        (1) Supervising maintenance (AKT2600) to include, but not limited to:
            (a) All round readings;
            (b) Agitation test;
            (c) Battery top up; and
            (d) Battery scrub.
        (2) Conducting milliamp test IAW C-26-509-000/MS-001;
        (3) Maintaining battery ventilation system (KT0690) by:
            (a) Visually inspecting submarine battery vent (YT0020) for:
                i. Dirty filters;
                ii. Pneumatic valves; and
                iii. Control & indication.
            (b) Testing battery vent airflow control units to confirm air flow readings within range.
        (4) Maintaining battery cooling system by:
            (a) Visually inspecting battery cooling fish plate (KT0160);
            (b) Visually inspecting battery cooling header pipe (KT0140);
            (c) Testing salinity;

(d) Removing ion exchange columns;
(e) Replacing ion exchange columns; and
(f) Sending ion exchange columns for refurbishing.

(5) Maintaining battery agitation by:
(a) Supervising battery agitation test before charge;
(b) Identifying defects; and
(c) Rectifying defects.

(6) Inspecting hydrogen eliminators (KT0200) by:
(a) Inspecting heaters & palladium envelopes for damage to include, but not limited to:
i. Dents; and
ii. Cracks.
(b) Checking electrical supplies (switches & lights).

b. Performing battery CM (AUT1960, AUT1970), if PM results are outside acceptable ranges (excluding battery scrub), IAW all refs and safety precautions by:

(1) Identifying symptoms of a sick battery cell by:
(a) Repairing defects IAW C-26-509-000/MS-001;
(b) Repairing adverse agitation readings; and
(c) Conducting burst bag routine.

(2) If not rectified, cutting out cell.

c. Updating battery Log (AUT4090) on completion of CMs and PMs; and

d. Analyzing battery log historical readings (AKT1080, AKT1090) to identify trends.

16. **Training Limitations**: Environment in training doesn't reflect limited space on submarines. Only 2 cells are available for training, 3 cell tops and cut-out devices are required for training.

EO 001.01 – BATTERY PM
LG 001.01.01 – BATTERY PM

LG 001.01.01 – Battery PM		
Location Conditions	**Instructional Sequence**	
Submarine Electrical Trainer/S-100 Classroom	Beginning of course	
Time	**Classification**	**Instructor**
450 minutes	Unclas	AIMM Qualified Guest Speaker
Learning Objectives	**Instructional Method(s)**	
1) PM scheduling 2) Reasoning for PMs 3) Fault finding 4) Safety precautions 5) Battery log historical trends	1) Guided Instruction 2) Guest Speaker 3) Individual Assignment 4) Group Discussion 5) Think, Pair, Share 6) Case Study 7) Battery Log Historical Trends Activity	
Scenario	While under contractor control, PM routines were not followed. This resulted in premature failure of the battery which forced the submarine to return to port 2 weeks into a 6-month deployment	

Activity Steps	Guided Instruction with PowerPoint augmentation: (100 minutes)
	• Battery chemistry & interactions
	• Battery capacity test
	• Battery PM schedule
	• Reasoning
	• Battery auxiliaries PM schedule
	• Charging routines
	• Identifying & rectifying defects
	• Iso-amplifiers/earth monitoring
	
	Guest Speaker: (if available member of FMFCS or LCMM) (100 minutes)
	• Battery log distribution
	• Battery degradation trending
	• Explain what is happening
	• Identify reasoning (why)
	
	Individual Assignment Navigating BR 2075: (Homework)
	• Answer questions regarding:
	  • Minimum permissible discharge voltage
	  • When can a D4 be omitted?
	  • How to get around D4?
	  • Mossing
	  • Limits of impurities
	  • Topping up – when and why
	  • (10 questions)	

LG 001.01.01 – Battery PM	

	Homework Review: (25 minutes)
	Group Discussion on Interpreting BR 2075: (25 minutes)
	• Where to access
	• Where to find scheduling
	• Reasoning for doing PMs
	
	Think-Pair Share: (50 minutes)
	Students will write down when PM should be done individually. They will then discuss their answers in pairs. Each pair will then share their views with the group. Instructor will add anything missed and make corrections as required.
	• Scheduling of PMs WRT:
	• All round readings
	• Agitation test
	• Battery top up
	• Battery scrub
	• Milliamp test
	• Battery ventilation system
	• Air flow readings within range
	• Battery cooling system
	  • Demineralized water
	  • Tank(s)
	  • Level range
	• Testing salinity
	• Battery agitation
	• Heaters & palladium envelopes
	• Ion exchange columns
	• Hydrogen eliminators
	
	Case Study: (50 minutes)
	Instructor will provide case study to students on when safety precautions weren't followed during PMs or on operations.
	• Equipment safety
	• Personnel safety
	
	Confirmation Activity Battery Log Historical Trends: (100 minutes)
	• Compare various HMCS Submarines at different battery life intervals
	• Identify trends
	• Explain what is happening
	• Identify reasoning (why)

Instructor Resources	**References**
1) PowerPoint 2) Guest speaker 3) BR 2075 handout 4) Safety case study	1) BR 2075 2) B1, B33, B53, B57, B58, B59, B62, B97, B114, B118, D1

LG 001.01.01 – Battery PM	
5) HMCS Subs battery log (at least 4)	
Note(s) to Instructor	
1) N/A	

EO 001.01 – BATTERY PM
EC 001.01 – BATTERY PM

EC 001.01 – Perform Battery PM		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.01	Submarine Electrical Trainer/S-100 Classroom	After LG 001.01.001
Time	**Classification**	**Instructor**
50 minutes	Unclass	AIMM Qualified
Scenario	You're on watch, the EO comes back aft and asks for information pertaining to the battery.	
Assessment Steps	**Input** – Student will be given written test on PM, battery faults, and briefing command. **Performance** – Student will answer questions on written test. **Output** – The trainee must achieve 10/14 SATs to pass this EC **Feedback** –From instructor to student individually following completion of assessment.	
Instructor Resources		
1) Written test 2) PP&S		
Note(s) to Instructor		
1) In Section 2 of checklist, "Explain why faults happen" of the many faults possible the trainee's explanation must include at least 3 of the following: agitation, cooling, burst bag, and sick cell.		
Assessment Limitations	Nil.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 001.01 BATTERY PM			
Item	**SAT**	**UNSAT**	**Remarks**
1. PM			
Listed charge routine	☐	☐	
Listed PM schedule	☐	☐	
Identified battery log distribution	☐	☐	
2. Identify Battery Faults (Fault 1)			
Identified faults for battery	☐	☐	
Substantiated reasoning for fault	☐	☐	
Explained why faults happen (3 identified)	☐	☐	
3. Identify Battery Faults (Fault 2)			
Identified faults for battery	☐	☐	
Substantiated reasoning for fault	☐	☐	
Explained why faults happen (3 identified)	☐	☐	
4. Identify Battery Faults (Fault 3)			
Identified faults for battery	☐	☐	
Substantiated reasoning for fault	☐	☐	
Explained why faults happen (3 identified)	☐	☐	
5. Brief Command			
Provide brief to command	☐	☐	
Brief was logical, correct and data interpreted properly	☐	☐	

Score: /14

The trainee must achieve 10/14 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee		

EC 001.01 BATTERY PM		
I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 001.02 – BATTERY CM
LG 001.02.01 – BATTERY CM

LG 001.02.01 – Battery CM		
Location Conditions	**Instructional Sequence**	
Submarine Electrical Trainer/S-100 Classroom	After EC 001.01	
Time	**Classification**	**Instructor**
100 minutes	Unclass	AIMM Qualified
Learning Objectives	**Instructional Method(s)**	
1) Battery CM 2) Burst bag routine 3) Cutting out a cell	1) EDI (in pairs)	
Scenario	While conducting ARR's, it was noticed that some of the cell's electrolyte temperatures were much higher than the temperatures in the remaining cells. Student will conduct corrective maintenance.	
Activity Steps	Explain, Demonstrate, & Imitate (in pairs): (100 minutes) • Explain/conduct: battery CM • Demonstrate & Imitate: Burst bag routine • Demonstrate & Imitate: Cutting out a cell	
Instructor Resources	**References**	
1) 3D printed cell tops/cell cut outs 2) 3D printed cut out kit 3) 3D printed torque wrench 4) Multi-meter (Fluke)	1) B1, B33, B53, B57, B58, B59, B62, B97, B98, B114, B118, D1	
Note(s) to Instructor		
1) N/A		

EO 001.02 – BATTERY CM
EC 001.02 – BATTERY CM

EC 001.02 – Battery CM		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.02	Submarine Electrical Trainer/S-100	After LG 001.02.01
Time	**Classification**	**Instructor**
150 minutes 25 minutes/ pair 25 minute debrief/ pair 3 x 50 min / 6 students	Unclass	AIMM Qualified

Scenario	You've received PM readings with symptoms of sick cell. After performing agitation test: Agitation was repaired prior to the charge, and it was found agitation wasn't the issue. After burst bag routine: Burst-bag routine verifies that cell is sick.
Assessment Steps	EC to be done in pairs, instructor will direct questions toward each student individually. **Input** – Student receives PM readings with symptoms of sick cell and is expected to explain how to guide a Mar Tech Jr through CM process, cut out sick cell, and answer follow-up question. **Performance** – orally explain how to verify agitation and other factors, conduct burst bag routine, findings show sick cell, cut out sick cell in pairs, and respond to follow-up question. **Output** – The trainee must achieve 6/7 SATs to pass this EC. **Feedback** – From instructor to group following completion of assessment.

Instructor Resources

1) Meter
2) PM readings
3) 3D printed cell tops/cell cut outs
4) 3D printed cut out kit (including cut out procedure)
5) 3D printed torque wrench
6) 3D printer wire

Note(s) to Instructor

1) Instructor to interject throughout EC with information from scenario following completion of each step.

Assessment Limitations	Incomplete battery cell on trainers. Space isn't restricted in the trainer whereas it is on the Submarine.

EC 001.02 – Battery CM	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

PROTECTED B (when complete)

EC 001.02 BATTERY CM			
Item	**SAT**	**UNSAT**	**Remarks**
1. Explanation of CM Process			
Explained how to verify battery agitation	☐	☐	
2. Performing Battery CMs			
Carried out burst bag routine	☐	☐	
Explained an attempted repair of adverse battery agitation readings	☐	☐	
3. Perform Cutting Out of Cell			
Safety precautions followed	☐	☐	**CRITICAL**
Followed procedure	☐	☐	**CRITICAL**
Used tools correctly	☐	☐	
4. Follow-up Question			
Responded to question regarding adverse conditions in battery (mid-cell, top-step, rubber mat, etc.)	☐	☐	

Score: /7

Trainee must achieve 4/5 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – MAINTAIN DIESEL GENERATOR

1. **Performance**: Maintain Diesel Generator.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Operational DG (diesel generator);
      (3) Electrical tools;
      (4) Degreaser or sonic bath;
      (5) PPE; and
      (6) Test equipment (including high voltage test lead).

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain Diesel Generator, IAW all refs, by:

   a. Maintaining diesel systems by:

      (1) Maintaining diesel auxiliaries by maintaining high voltage precipitator (KT0150); and
      (2) Troubleshooting main diesel engine start control to include, but not limited to:
         (a) Speed probe (AUT1460) by:
            i. Visually inspecting;
            ii. Checking cable tightness;
            iii. Performing a function test; and
            iv. Replacing if required.
         (b) Solenoids (AUT1450) by:
            i. Visually inspecting;
            ii. Checking cable tightness;
            iii. Performing a function test; and
            iv. Replacing if required.

   b. Maintaining generator by:

      (1) Testing generator rectifier (KT0520) by:
         (a) Inspecting generator rectifier set (KT0190);

(b) Reading generator break down readings (inspecting generator fuses);
(c) Identifying issue with rectifier; and
(d) Repairing generator rectifier (KT0360).

4. **Training Limitations**: Must be carried out on submarine platform as no trainer is available.

EO 002.01 – MAINTAIN DIESEL SYSTEM
LG 002.01.01 – MAINTAIN DIESEL AUXILIARIES

LG 002.01.01 – Maintain Diesel Auxiliaries		
Location Conditions	**Instructional Sequence**	
Submarine Diesel Trainer / S100 Classroom	After EC 001.02	
Time	**Classification**	**Instructor**
150 minutes	Unclass	AIMM Qualified
Learning Objectives	**Instructional Method(s)**	
1) Maintain diesel auxiliary systems	1) EDI	
Scenario	You are tasked to complete the PM on both port and stbd diesel precipitators.	
Activity Steps	Explain, Demonstrate, & Imitate the following: • maintaining high voltage precipitator – location of precipitator isolation • discharge capacitor with correct equipment (attitude - safety/damage equipment/injury) • clean precipitator fins • Maintaining diesel auxiliaries by maintaining high voltage precipitator	
Instructor Resources	**References**	
1) Sonic bath 2) Degreaser(non-conductive) 3) 10 mil wrench 4) Multimeter with 10kV probe 5) Rags	1) B1, B33, B48, B49, B50, B51, B78, B80, B117	
Note(s) to Instructor		
1) N/A		

EO 002.01 – MAINTAIN DIESEL SYSTEM
LG 002.01.02 – SOLENOIDS AND SPEED PROBE

LG 002.01.02 – Solenoids and Speed Probe		
Location Conditions	**Instructional Sequence**	
Submarine Diesel Trainer / S100 Classroom	After LG 002.01.01	
Time	**Classification**	**Instructor**
`150 minutes	Unclass	AIMM Qualified
Learning Objectives	**Method(s)**	
1) Location on subs 2) Inspection 3) Replace	1) EDI	
Scenario	You are tasked to pull, clean, inspect, and reinstall speed solenoid and back up solenoid. Also ensure internal gear is correctly aligned.	
Activity Steps	Note: Individuals will have knowledge of basic speed probes and solenoids, must learn location and common faults. Testing can't be practical, due to no trainer and having on OJPR would increase wear and tear on equipment. Also, not difficult. Activity: Instructor to provide reading outside of specifications range and have them verbally identify causes and walk-through repair procedures: 1) Troubleshooting main diesel engine start control to include, but not limited to the following: a) Speed probe by; i. Visually inspecting; ii. Checking cable tightness; iii. Performing a function test; and iv. Replacing if required. b) Solenoids by, i. Visually inspecting; ii. Checking cable tightness; iii. Performing a function test; and iv. Replacing if required.	
Instructor Resources	**References**	
1) Appropriate wrenches 2) Appropriate screw drivers 3) Degreaser (non-conductive)	1) B1, B33, B48, B49, B50, B51, B78, B80, B117	
Note(s) to Instructor		
1) N/A		

EO 002.01 – MAINTAIN DIESEL SYSTEM
EC 002.01 – MAINTAIN DIESEL SYSTEM

EC 002.01 – Maintain Diesel System		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.01	Submarine Diesel Trainer / S100 Classroom	Day after LG 002.01.02
Time	**Classification**	**Instructor**
300 minutes (1 x 50 min period per student)	Unclass	AIMM Qualified
Scenario	You are tasked to complete the PM on both port and stbd diesels.	
Assessment Steps	**Input** – **Performance** – Hands on EC. **Output** – Trainee must achieve 7/9 SATs on non-critical factors AND all critical factors to pass this EC. **Feedback** – From instructor to student individually following completion of assessment.	
Instructor Resources		
1) Diagram of generator cut away 2) PP&S		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	EC to be conducted individually.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 002.01 MAINTAIN DIESEL SYSTEM			
Item	**SAT**	**UNSAT**	**Remarks**
1. Maintain Diesel Auxiliaries			
Identify location of precipitator isolation	☐	☐	
Use appropriate PPE to Safely discharge capacitor	☐	☐	**CRITICAL**
Safely discharge capacitor	☐	☐	
Clean precipitator fins	☐	☐	
2. Solenoids and Speed Probe			
Inspect Solenoids	☐	☐	
Perform function test on Solenoids	☐	☐	
Replace Solenoid	☐	☐	
Inspect Speed Probe	☐	☐	
Perform function test on Speed Probe	☐	☐	
Replace Speed Probe	☐	☐	

Score: /10

Trainee must achieve 7/9 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 002.02 – MAINTAIN DIESEL GENERATOR
LG 002.02.01 – MAINTAIN DIESEL GENERATOR

LG 002.02.01 – Maintain Diesel Generator		
Location Conditions	**Instructional Sequence**	
Submarine Diesel Trainer / S100 Classroom	After EC 002.01	
Time	**Classification**	**Instructor**
200 minutes	Unclas	AIMM Qualified
Learning Objectives	**Instructional Method(s)**	
1) Generator functionalities 2) AC motor 3) Rectifier 4) Personnel safety points 5) Equipment safety points 6) Lead placement 7) Lead readings 8) Diode procedure	1) Guided Instruction 2) Think, Pair, Share 3) Diagram Labelling 4) Group Discussion	
Scenario	You have been tasked with inspecting the rotating rectifier assembly.	

Activity Steps	Guided Instruction with PowerPoint augmentation: (50 minutes)
	• Generator functionalities (poles, windings, facts, stats)
	• AC motor
	• Rectifier (ramifications of incorrect diode placement)
	
	Think, Pair, Share: (20 minutes)
	Students will write down what they believe are the personnel and equipment safety considerations individually. They will then discuss their answers in pairs. Each pair will then share their views with the group. Instructor will add anything missed and make corrections as required.
	• Personnel safety points:
	  • Meter check
	  • Isolate diesel engine
	  • Ensure generator breaker is open
	  • Pull fuses
	• Equipment safety points:
	  • Pull fuses
	  • Proper replacement of diode
	
	Diagram Labelling: (20 minutes)
	Instructor will show students lead placement through the use of a diagram.
	• Lead placement
	Group Discussion: (15 minutes)
	Instructor will lead a discussion pertaining to what to expect for readings on various diagrams.
	• What to expect for lead readings
	
	Guided Instruction with PowerPoint augmentation: (20 minutes)	

LG 002.02.01 – Maintain Diesel Generator	
• Faulty diode procedure Confirmation Activity with Discussion: (50 minutes) • Diagram labelling with leads and explanations	
Instructor Resources	**References**
1) PowerPoint 2) Personnel safety points solutions 3) Equipment safety points solutions 4) Diagram with lead placement 5) Various diagrams with differing lead readings 6) Diagram with leads and explanations	1) B1, B33, B48, B49, B50, B51, B78, B80, B117
Note(s) to Instructor	
1) N/A	

EO 002.02 – MAINTAIN DIESEL GENERATOR
EC 002.02 – MAINTAIN DIESEL GENERATOR

EC 002.02 – Maintain Diesel Generator		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.02	Submarine Diesel Trainer / S100 Classroom	Day After LG 002.02.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Scenario	You have been tasked with repairing the rotating rectifier assembly.	
Assessment Steps	**Input** – Question at beginning for safety, diagram of generator cut away with leads identified (6), space underneath diagram for explanations, and question at end on why a diode would be replaced and the procedure to do so. **Performance** – Identify personnel and equipment safety, label leads positive or negative, identify correct meter readings, explain all identified lines, and explain replacement of faulty diode procedure and why it would be conducted. **Output** – The trainee must achieve 5/6 SATs to pass this EC. **Feedback** – From instructor to student individually following completion of assessment.	
Instructor Resources		
1) Diagram of generator cut away 2) PP&S		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	EC to be conducted individually.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 002.02 MAINTAIN DIESEL GENERATOR			
Item	**SAT**	**UNSAT**	**Remarks**
1. Label Diagram of Generator Cut Away			
Personnel safety points identified (minimum 3)	☐	☐	**CRITICAL**
Equipment safety points identified (minimum 2)	☐	☐	
Lines for leads identified (positive or negative) for each line on the diagram	☐	☐	
Meter readings identified	☐	☐	
2. Steps Explained			
Each identified line explained	☐	☐	
Replacement of faulty diode procedure explained	☐	☐	

Score: /6

Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this EC.
The trainee must achieve 5/6 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 003 – MAINTAIN DC POWER DISTRIBUTION

1. **Performance**: Maintain DC Power Distribution.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Operational DC distribution system;
      (3) Electrical tools;
      (4) PPE;
      (5) DRMIS;
      (6) Safety member; and
      (7) Test equipment.

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine, and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain DC Power Distribution, IAW all refs and DRMIS, by:

   a. Printing out maintenance routines from DRMIS (AKT1250);

   b. Maintaining DC motors and starters by:

      (1) Troubleshooting DC starters by:
         (a) Testing DC starters (KT0500);
         (b) Inspecting DC starters;
         (c) Identifying faults with DC starters; and
         (d) Repairing faults if applicable.
      (2) Conducting PM on DC motor to include, but not limited to:
         (a) Performing inspection to include:
            i. Checking brushes (AKT0800);
            ii. Checking commutator; and
            iii. Performing neutral plane adjustment.
         (b) Performing insulation resistance checks (AT0070).
      (3) Maintaining water sensitive probe to include:
         (a) Inspecting water sensitive probe; and
         (b) Cleaning water sensitive probe.

c. Inspecting CO2 absorption unit (KT0180) by:

(1) Troubleshooting CO2 absorption unit (KT0710) by:
(a) Performing a function test on CO2 absorption unit.

d. Conducting main power bus readings (AT0070) by:

(1) Isolating power for breakdown readings;
(2) Performing readings;
(3) Analyzing readings;
(4) Identifying faults; and
(5) Repairing faults if required.

e. Maintaining DC load centres by:

(1) Performing PMs on DC load centres by:
(a) Repairing or replacing:
i. PSU (power supply units),
ii. Filters, and
iii. Overload relay units.
(b) Ordering parts if required.

f. Logging completed maintenance in DRMIS (AFT0400).

4. **Training Limitations**: There is no trainer available for this, a submarine platform is required during the directed training portion.

EO 003.01 – DC MOTORS AND STARTERS
LG 003.01.01 – DC MOTORS AND STARTERS

LG 003.01.01 – DC MOTORS AND STARTERS		
Location Conditions	**Instructional Sequence**	
Submarine Diesel Trainer / S100 Classroom	After EC 002.02	
Time	**Classification**	**Instructor**
350 minutes	Unclas	AIMM Qualified
Learning Objectives	**Instructional Method(s)**	
1) PM on DC motor 2) PM & troubleshooting on DC starter 3) Personnel safety requirements 4) Equipment safety requirements 5) Permissible IR readings 6) Water sensitive probe maintenance procedure 7) CO2 absorption unit	1) Group Discussion 2) Scenarios 3) Think, Pair, Share 4) Guided Instruction 5) Guided Discussion 6) Video	
Scenario	You have been tasked with conducting PM on the HP Bilge Pp and starter.	
Activity Steps	Group Discussion: (100 minutes) *Students will receive PM on DC motor and starter then read individually and discuss as a group. While this discussion takes place DC motor/starter is on screen.* • PM on DC motor • Components & normal voltage • Operational sequence of DC starter Scenarios: (100 minutes) *Instructor will give examples on common faults.* • DC motor/starter faults to include resistance and voltage readings and visible defects. ▪ Faulty contactor ▪ Faulty PSU ▪ Faulty relay Think, Pair, Share: (50 minutes) *Students will write down what they believe are the personnel and equipment safety considerations individually. They will then discuss their answers in pairs. Each pair will then share their views with the group. Instructor will add anything missed and make corrections as required.* • Live troubleshooting ▪ Personnel safety requirements ▪ 2nd person with cane ▪ Gloves ▪ Equipment Safety Requirements ▪ Short circuiting • De-energized troubleshooting ▪ Personnel safety requirements	

LG 003.01.01 – DC MOTORS AND STARTERS	
	▪ Isolate power ▪ Test for power ▪ Discharge filter ▪ Equipment Safety Requirements ▪ Insulation resistance readings/meggering Guided Instruction with PowerPoint augmentation: (25 minutes) • Permissible IR reading ranges • Procedure if IR reading out of range • Causes of IR reading degradation Guided Discussion: (25 minutes) • Water sensitive probe maintenance procedure ▪ Rundown of PM ▪ Potential defects Guided Discussion: (25 minutes) • CO2 absorption unit • Function test IAW SOP Confirmation Activity: (25 minutes) *Video of DC starter with numbering pop-up to label what is happening at each step. Students will identify what is happening at each step. Instructor will host discussion to confirm knowledge.*
Instructor Resources	**References**
1) DRMIS print out of PM on DC motor 2) 3D rendering of DC motor 3) 3D rendering of DC starter 4) List of components & normal voltages 5) Video of DC starter 6) Computer 7) Screen	1) B1, B33, B43, B44, B45, B46, B47, B52, B53, B65, B80, B118, B120
Note(s) to Instructor	
1) VR is pending for AC/DC starter, check with NTDC(A) prior to running course.	

EO 003.01 – DC MOTORS AND STARTERS
EC 003.01 – DC MOTORS AND STARTERS

EC 003.01 – DC Motors and Starters		
Scope	**Location Conditions**	**Instructional Sequence**
EO 003.01	Submarine Diesel Trainer / S100 Classroom	Day after LG 003.01.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Scenario	You have been tasked with conducting PM on the HP Ballast Pp and starter.	
Assessment Steps	**Input** – Written test with questions as follows: personnel safety, equipment safety, minimum permissible IR readings, diagram to label components of DC starter, and steps in water sensitive probe maintenance procedure. **Performance** – Answer all questions with minimum answers as indicated on EC checklist, and complete diagram labelling. **Output** – The trainee must achieve 6/8 to pass this EC. **Feedback** – From instructor to student individually following completion of assessment.	
Instructor Resources		
1) DRMIS maintenance routines 2) Written test 3) PP&S 4) Diagram		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	Nil.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

EC 003.01 DC MOTORS AND STARTERS			
Item	**SAT**	**UNSAT**	**Remarks**
1. Conduct PM on DC Motor			
Identify personnel safety requirements (minimum 3)	☐	☐	**CRITICAL**
Identify equipment safety requirements (minimum 2)	☐	☐	**CRITICAL**
Identify minimum permissible IR readings	☐	☐	
2. Conduct PM & Troubleshoot DC Starter			
Identify all components of DC starter (minimum 70% of components labelled)	☐	☐	
List checks for starter (minimum 5)	☐	☐	
3. Maintain Water Sensitive Probe			
Identify personnel safety requirements (minimum 1)	☐	☐	
Identify equipment safety requirements (minimum 2)	☐	☐	
Identify steps in water sensitive probe maintenance procedure (minimum 3 steps)	☐	☐	

Score: /8

Trainee must achieve 4/6 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

CO NTDC(A) Approval – 04/07/2022
CNPTG Approval – 21/08/2022
Version No: 3.0

EO 003.02 – MAIN POWER BUS READINGS
LG 003.02.01 – MAIN POWER BUS READINGS

LG 003.02.01 – Main Power Bus Readings		
Location Conditions	**Instructional Sequence**	
Submarine Electrical Trainer /S-100 Classroom	After LG 003.01.01	
Time	**Classification**	**Instructor**
100 minutes	Unclas	AIMM Qualified
Learning Objectives	**Instructional Method(s)**	
1) Main power bus readings 2) Safety precautions 3) Identifying faults	1) Group Discussion 2) Scenarios	
Scenario	You have been instructed to gather readings on the main power bus.	
Activity Steps	Guided Instruction with PowerPoint augmentation: (35 minutes) • Main power bus readings • Safety precautions (including lock out) Scenarios: (15 minutes) *Instructor will give examples on common faults.* • Identifying faults • Possible solutions Trainer: (25 minutes) *Instructor will bring students to see propulsion switchboard trainer with covers removed.*	
Instructor Resources	**References**	
1) PowerPoint with steps and circuits 2) Scenarios	1) B1, B33, B43, B44, B45, B46, B47, B52, B53, B65, B80, B118, B120	
Note(s) to Instructor		
1) N/A		

EO 003.02 – MAIN POWER BUS READINGS
EC 003.02 – MAIN POWER BUS READINGS

EC 003.02 – Main Power Bus Readings		
Scope	**Location Conditions**	**Instructional Sequence**
EO 003.02	Submarine Electrical Trainer/S-100 Classroom	Day after LG 003.02.01
Time	**Classification**	**Instructor**
100 minutes	Unclas	AIMM Qualified
Scenario	You have been instructed to gather readings on the main power bus.	
Assessment Steps	**Input** – Oral test with questions as follows: personnel safety, equipment safety, and minimum permissible IR readings. **Performance** – Respond to all questions with minimum answers as indicated on EC checklist. **Output** – The trainee must achieve 3/3 SATs to pass this EC. **Feedback** – From instructor to as a group following completion of assessment.	
Instructor Resources		
1) DRMIS maintenance routines		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	Nil.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 003.02 MAIN POWER BUS READINGS			
Item	**SAT**	**UNSAT**	**Remarks**
1. Main Power Bus Readings			
Identify personnel safety requirements (minimum 3)	☐	☐	**CRITICAL**
Identify equipment safety requirements (minimum 2)	☐	☐	**CRITICAL**
Identify minimum permissible IR readings	☐	☐	

Score: /3

Trainee must achieve 1/1 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 003.03 – MAINTAIN DC LOAD CENTRES
LG 003.03.01 – MAINTAIN DC LOAD CENTRES

LG 003.03.01 – Maintain DC Load Centres		
Location Conditions	**Instructional Sequence**	
Submarine Electrical Trainer /S-100 Classroom	After LG 003.02.01	
Time	**Classification**	**Instructor**
100 minutes	Unclas	AIMM Qualified
Learning Objectives	**Instructional Method(s)**	
1) PMs on DC load centers 2) Safety precautions	1) Guided Instruction 2) Group Discussion 3) Scenarios 4) Think, Pair, Share	
Scenario	You have been tasked with conducting PM on the Fwd DCLC.	
Activity Steps	Guided Instruction with PowerPoint augmentation: (35 minutes) • PMs on DC load centers • Components of DC load centers • Safety precautions Group Discussion: (30 minutes) *Students will receive PM on DC load centre, read individually, and discuss as a group. While this discussion takes place DC load centre diagram on screen.* • PM on DC load centres • Components & normal voltage Think, Pair, Share: (30 minutes) *Students will write down what they believe are the personnel and equipment safety considerations individually. They will then discuss their answers in pairs. Each pair will then share their views with the group. Instructor will add anything missed and make corrections as required.* • Live troubleshooting ▪ Personnel safety requirements ▪ 2nd person with cane ▪ Gloves ▪ Equipment Safety Requirements ▪ Short circuiting • De-energized troubleshooting ▪ Personnel safety requirements ▪ Isolate power ▪ Test for power ▪ Discharge filter ▪ Equipment Safety Requirements ▪ Insulation resistance readings/meggering Scenarios: (15 minutes) *Instructor will give examples on common faults.*		

LG 003.03.01 – Maintain DC Load Centres	
	• Identifying faults (e.g., PSU failure, remote) • Possible solutions
Instructor Resources	**References**
1) PowerPoint with steps and circuits 2) Scenarios 3) Computer 4) Screen 5) Scenario with common faults	1) B1, B33, B43, B44, B45, B46, B47, B52, B53, B65, B80, B118, B120
Note(s) to Instructor	
1) Possible changes based on VR availability, speak with NTDC(A) staff prior to running course.	

EO 003.03 – MAINTAIN DC LOAD CENTRES
EC 003.03 – MAINTAIN DC LOAD CENTRES

EC 003.03 – Maintain DC Load Centres		
Scope	**Location Conditions**	**Instructional Sequence**
EO 003.03.01	Submarine Electrical Trainer /S-100 Classroom	Day after LG 003.03.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Scenario	You have been tasked with conducting PM on the Aft DCLC.	
Assessment Steps	**Input** – Written test with questions as follows: personnel safety, equipment safety, and diagram to label components of DC load centre. **Performance** – Answer all questions with minimum answers as indicated on EC checklist, and complete diagram labelling. **Output** – The trainee must achieve 3/3 SATs to pass this EC. **Feedback** – From instructor to student individually following completion of assessment.	
Instructor Resources		
1) Diagram of DC load centre 2) Written test 3) PP&S		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	Nil.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 003.03 MAINTAIN DC LOAD CENTRES			
Item	**SAT**	**UNSAT**	**Remarks**
1. Safety Precautions			
Identify personnel safety requirements (minimum 3)	☐	☐	**CRITICAL**
Identify equipment safety requirements (minimum 2)	☐	☐	**CRITICAL**
2. Maintain DC Load Centre			
Identify all components of DC load centre (minimum 70% of components labelled)	☐	☐	

Score: /3

Trainee must achieve 1/1 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 004 – MAINTAIN MAIN PROPULSION SYSTEM

1. **Performance**: Maintain Main Propulsion System.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Main propulsion system;
      (3) PPE;
      (4) Safety member;
      (5) Electrical tools; and
      (6) Test equipment (including Insulation Resistance Tester).

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain Main Propulsion System by:

   a. Inspecting field supply converters (BAT0090, KT0510), IAW maintenance manual, to include but not limited to:

      (1) Cleanliness;
      (2) Cabling;
      (3) Connectivity;
      (4) Fuses; and
      (5) Outputs.

   b. Conducting main motor breakdown readings (BT0260) by:

      (1) Participating as a team member;
      (2) Isolating power by performing lock out, tag out IAW refs; and
      (3) Cataloguing readings.

   c. Maintaining ACU (Automatic Control Unit) (BT0220) by:

      (1) Troubleshooting ACU by:
         (a) Reading code card;
         (b) Identifying issues IAW BR1966 (9) 12;

(c) Performing equipment safety functions for static electricity (i.e. straps);
(d) Calibrating circuit cards;
(e) Adjusting circuit card connections;
(f) Inserting card; and
(g) Performing function test.

4. **Training Limitations**: Main motor breakdown readings are 3rd line maintenance and students may not have the opportunity to practice on OJPR. The main propulsion system trainer doesn't simulate, it will only show test points.

EO 004.01 – FIELD SUPPLY CONVERTERS
LG 004.01.01 – FIELD SUPPLY CONVERTERS

LG 004.01.01 – Field Supply Converters		
Location Conditions		**Instructional Sequence**
Submarine Electrical Training Facility/S-100 Classroom		After LG 003.03.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) PMs 2) Inspection		1) Group Discussion
Scenario	You have been tasked with conducting PM on the Field Supply Converters.	
Activity Steps	Group Discussion: (50 minutes) *Students will receive PM on field supply converters and discuss as a group. While this discussion takes place with field supply converters on screen.* • PM on field supply converters • Inspecting	
Instructor Resources		**References**
1) DRMIS maintenance routines		1) B1, B2, B3, B32, B47, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B117, B120
Note(s) to Instructor		
1) N/A		

EO 004.02 – MAIN MOTOR BREAKDOWN READINGS
LG 004.02.01 – MAIN MOTOR BREAKDOWN READINGS

LG 004.02.01 – Main Motor Breakdown Readings		
Location Conditions		**Instructional Sequence**
Submarine Electrical Training Facility/S-100 Classroom		After LG 004.01.01
Time	**Classification**	**Instructor**
400 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) Procedures 2) Parameters		1) EDI
Scenario	Class will work as a team, led by the instructor, to carry out main motor breakdown readings.	
Activity Steps	Conducting main motor breakdown readings by: 1) Participating as a team member; 2) Isolating power by performing lock out, tag out IAW refs; and 3) Cataloguing readings.	
Instructor Resources		**References**
1) Instructions 2) Wiring harness(may need to make) 3) Multimeter 4) Rubber matting		1) B1, B2, B3, B32, B47, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B117, B120
Note(s) to Instructor		
1) N/A		

EO 004.03 – TROUBLESHOOT ACU
LG 004.03.01 – TROUBLESHOOT ACU

LG 004.03.01 – Troubleshoot ACU		
Location Conditions		**Instructional Sequence**
Propulsion Trainer/S-100 Classroom		After LG 004.02.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Learning Objectives		**Method(s)**
1) Card functions 2) Error codes (issue identification) 3) Calibrate cards 4) Relationship to field supply converters		1) Guided Discussion in Propulsion Trainer
Scenario	You have been tasked with conducting CM on the ACU.	
Activity Steps	Guided Discussion in Propulsion Trainer: (50 minutes) • Card functions & error codes • Calibrating cards (demonstration) • Inserting cards • Relationship between field supply converters and propulsion switchboard ▪ Selection switches	
Instructor Resources		**References**
1) Propulsion trainer 2) Cards		1) B1, B2, B3, B32, B47, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B117, B120
Note(s) to Instructor		
1) N/A		

EO 004.03 – TROUBLESHOOT ACU
EC 004.03 – FIELD SUPPLY CONVERTERS & TROUBLESHOOTING ACU

EC 004.03 – Title		
Scope	**Location Conditions**	**Instructional Sequence**
EO 004.01 & EO 004.03	Propulsion Trainer/S-100 Classroom	Day after LG 004.03.01
Time	**Classification**	**Instructor**
200 minutes	Unclas	AIMM Qualified
Scenario	You have been tasked with conducting PM and CM on the Main Motor.	
Assessment Steps	**Input** – Candidates will swap card, perform a functions test in the ACU trainer, and respond to field supply converters question. **Performance** – Candidates will identify a faulty card, remove said card, calibrate a replacement card, install said card, perform a function test, and respond to field supply converters question. **Output** – The trainee must achieve 4/6 SATs to pass this EC. **Feedback** – From instructor to group following completion of assessment.	
Instructor Resources		
1) ACU trainer 2) Faulty card 3) Replacement card 4) Pre-Seas SOP 5) Calibration tools 6) ACU manual		
Note(s) to Instructor		
1) Put error code for fault into trainer.		
Assessment Limitations	Nil	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 004.03 FIELD SUPPLY CONVERTERS & TROUBLESHOOTING ACU			
Item	**SAT**	**UNSAT**	**Remarks**
1. Swap Card			
Identify faulty card	☐	☐	
Remove faulty card	☐	☐	
Calibrate replacement card	☐	☐	
Install replacement card	☐	☐	
2. Perform Function Test			
Perform function test IAW Pre-Seas SOP	☐	☐	
3. Field Supply Converters			
Responded to field supply converters question (e.g. PM)	☐	☐	

Score: /6

The trainee must achieve 4/6 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 005 – MAINTAIN VC-AS

1. **Performance**: Maintain VC-AS.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Operational VC-AS (Victoria Class Autopilot System);
      (3) POU (portable operating unit);
      (4) Electrical tools;
      (5) All operational hydraulics systems;
      (6) RSU (ram servo unit);
      (7) Planing surfaces; and
      (8) Test equipment.

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain VC-AS by:

   a. Operating VC-AC System (BAT0160) by:

      (1) Ensuring planing surfaces are clear of obstructions and personnel (i.e. divers down).

   b. Making appropriate warning pipes.

   c. Performing troubleshooting function test by:

      (1) Testing port & starboard computers (KT0540) by:
         (a) Energizing system;
         (b) Testing position indicators (KT0550);
         (c) Inspecting ram servo linkage (KT0220); and
         (d) Performing bite test.
      (2) Troubleshooting VC-AS (BAT0380) by:
         (a) Identifying defects; and
         (b) Repairing defects.

4. **Training Limitations**: Cannot simulate hydraulics system or perform bite test in trainer.

EO 005.01 – MAINTAIN VC-AS
LG 005.01.01 – OPERATE VC-AS

LG 005.01.01 – Operate VC-AS		
Location Conditions		**Instructional Sequence**
VC-AS Trainer/S100 Classroom		After LG 004.03.01
Time	**Classification**	**Instructor**
800 minutes	Unclassified	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) VC-AS Equipment Overview 2) VC-AS Functional Overview 3) Using the VC-AS Software 4) Operating the VC-AS		1) Interactive Lecture 2) Practical Demonstration
Scenario	The submarine is preparing to conduct pre seas on the VC-AS.	
Activity Steps	1) <u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class with all the required reference and instructional material. 2) <u>Introduction</u> (10 min) The instructor will explain to the students the importance of the VC-AS and the differences between the new system and the old system. 3) <u>Lecture</u> (750 min) Instructor shall discuss the importance of each of the learning objectives; specifically, why the VC-AS maintainer must have both an understanding of operation as it will assist with future troubleshooting. 4) <u>Guided Discussion</u> (30 min) After instruction, the instructor will ask students what the main takeaway was and if it changed their perspective on the importance the VC-AS, and why. 5) <u>Conclusion</u> (5 min) Ask if there are any questions.	
Instructor Resources		**References**
1) VC-AS trainer 2) Pre-Seas SOP		1) B11, B12, B13, B14, B15, B33, B37, B38, B39, B40, B64, B65, B67, B70, B117, B118, D2, D3, D4, D5, D6
Note(s) to Instructor		
1) Students should be familiar with each of the learning objectives by end of lectures.		

EO 005.01 – MAINTAIN VC-AS
LG 005.01.02 – TROUBLESHOOT VC-AS

LG 005.01.02 – Troubleshoot VC-AS		
Location Conditions		**Instructional Sequence**
VC-AS Trainer/S100 Classroom		After EO 005.01.01
Time	**Classification**	**Instructor**
300 minutes	Unclas	AIMM Qualified
Learning Objectives		**Method(s)**
1) VC-AS Equipment Overview 2) VC-AS LRU Replacement 3) VC-AS SW Tools 4) VC-AS HW Tools 5) VC-AS Trouble Shooting 6) VC-AS Maintenance Tasks		1) Interactive Lecture 2) Practical Demonstration
Scenario	The submarine has encountered a defect with the VC-AS.	
Activity Steps	1) <u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class with all the required reference and instructional material. 2) <u>Introduction</u> (10 min) The instructor will explain to the students the importance of the VC-AS and why it was updated. 3) <u>Lecture</u> (750 min) Instructor shall discuss the importance of each of the learning objectives; specifically, why the VC-AS maintainer must have both an understanding of operation and signal flow. 4) <u>Guided Discussion</u> (30 min) After instruction, the instructor will ask students what the main takeaway was and if it changed their prospective on the importance the VC-AS, and why. 5) <u>Conclusion</u> (5 min) Ask if there are any questions.	
Instructor Resources		**References**
1) VC-AS trainer		1) B11, B12, B13, B14, B15, B33, B37, B38, B39, B40, B64, B65, B67, B70, B117, B118, D2, D3, D4, D5, D6
Note(s) to Instructor		
1) Students should be familiar with each of the learning objectives by end of lectures.		

EO 005.01 – MAINTAIN VC-AS
EC 005.01 – MAINTAIN VC-AS

EC 005.01 – MAINTAIN VC-AS		
Scope	**Location Conditions**	**Instructional Sequence**
EO 005.01.01 & EO 005.01.02	VC-AS Trainer/S100 Classroom	Day after LG 005.01.02
Time	**Classification**	**Instructor**
50 minutes/student	Unclassified	AIMM Qualified

Scenario	The submarine has encountered a defect with the VC-AS.
Assessment Steps	Instructions Given to Trainee: 1) EC 005 will be a practical test. **Input** – Candidates will perform a functions test in the VC-AS trainer, and respond to faults found according to teachings in this PO. **Performance** – Candidates will identify faults, rectify faults and perform a function tests to ensure operational ability. **Output** – Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this EC. **Feedback** – From instructor to group following completion of assessment.

Instructor Resources

1) Classroom
2) Exam
3) All reference material
4) DS Solution

Note(s) to Instructor

1) On completion the instructor will inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Assessment Limitations	Nil.
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

PROTECTED B (when complete)

EC 005.01 MAINTAIN VC-AS			
Item	**SAT**	**UNSAT**	**Remarks**
1. Operate VC-AS			
Ensure surfaces are clear of obstructions,	☐	☐	**CRITICAL**
Ensure surfaces are clear of personnel, and	☐	☐	**CRITICAL**
Make appropriate warning pipes	☐	☐	
Operate VC-AS	☐	☐	
2. Troubleshoot VC-AS			
Function Tests Port and Stbd systems,	☐	☐	
Identify defects, and	☐	☐	
Repair defects.	☐	☐	
Prove System operational	☐	☐	
	☐	☐	

Score: /8

Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 005 – MAINTAIN VC-AS
PC 005 – MAINTAIN VC-AS

PC 005 – Maintain VC-AS		
Scope	**Location Conditions**	**Instructional Sequence**
PO 005	VC-AS Trainer/S100 Classroom	Day after EC 005.01
Time	**Classification**	**Instructor**
50 Minutes/student	Unclassified	AIMM Qualified
Scenario	The submarine has encountered a defect with the VC-AS.	
Assessment Steps	Instructions Given to Trainee: 2) PC 005 will be a practical test based on content from PO 005. **Input** – Candidates will perform a functions test in the VC-AS trainer, and respond to faults found according to teachings in this PO. **Performance** – Candidates will identify faults, rectify faults and perform a function tests to ensure operational ability. **Output** – Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this PC. 3) **Feedback** – From instructor to group following completion of assessment.	
Instructor Resources		
1) Classroom 2) Exam 3) All reference material 4) DS Solution		
Note(s) to Instructor		
1) Nil.		
Assessment Limitations	Nil	
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.	

PROTECTED B (when complete)
PROTECTED B (when complete)

PC 005 MAINTAIN VC-AS			
Item	**SAT**	**UNSAT**	**Remarks**
1. Operate VC-AS			
Ensure surfaces are clear of obstructions,	☐	☐	**CRITICAL**
Ensure surfaces are clear of personnel, and	☐	☐	**CRITICAL**
Make appropriate warning pipes	☐	☐	
Operate VC-AS	☐	☐	
2. Troubleshoot VC-AS			
Function Tests Port and Stbd systems,	☐	☐	
Identify defects, and	☐	☐	
Repair defects.	☐	☐	
Prove System operational	☐	☐	
	☐	☐	

Score: /8

Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PO 006 – MAINTAIN AC POWER GENERATION & DISTRIBUTION

1. **Performance**: Maintain AC Power Generation & Distribution.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Operational AC (alternating current) electrical system;
      (3) Electrical tools;
      (4) PPE;
      (5) Safety member; and
      (6) Test equipment.

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain AC Power Generation & Distribution by:

   a. Donning PPE.

   b. Maintaining 140kW MG by:

      (1) Inspecting PMG;
      (2) Inspecting commutator;
      (3) Conducting insulation resistance checks (AT0070);
      (4) Conducting function test (AUT3170); and
      (5) Supervising maintenance (AKT2600).

   c. Maintaining AC motors and starters by:

      (1) Troubleshooting hydraulic plant (KT0620) by:
         (a) Calibrating electro-hydraulic (Sirco) control valve.
      (2) Maintaining auto lubrication control system by:
         (a) Testing functionality in control box, if fail then:
            i. Troubleshooting auto lubrication control system (KT0560); and
            ii. Repairing auto lubrication control system.
      (3) Maintaining VCP (ventilation control panel) & associated equipment by:
         (a) Maintaining duct heaters by:

i. Testing duct heaters, if fail then:
1) Troubleshooting duct heaters; and
2) Repairing duct heaters.

(b) Maintaining ship's ventilation fans by:
i. Inspecting ventilation fans; and
ii. Testing ventilation fans, if fail then:
1) Troubleshooting ship's ventilation systems (KT0680); and
2) Repairing vent fans.

d. Maintaining SFCs (Static Frequency Converter) (AUT1470) by:

(1) Troubleshooting SFCs;
(2) Identifying defects;
(3) Replacing components; and
(4) Ordering parts as required.

e. Maintaining TRUs (Transformer Rectifier Unit) by:

(1) Load balancing TRUs;
(2) Troubleshooting TRUs;
(3) Identifying defects;
(4) Replacing components (AUT2850); and
(5) Ordering parts as required.

4. **Training Limitations**: Only available on submarine platform, no trainer is available.

EO 006.01 – 140KW MG
LG 006.01.01 – 140KW MG

LG 006.01.01 – 140kW MG		
Location Conditions		**Instructional Sequence**
S100 Classroom		After LG 005.01.02
Time	**Classification**	**Instructor**
250 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) Safety precautions 2) PMG 3) Commutator 4) Insulation resistance checks		1) Group Discussion 2) Video 3) Guided Instruction 4) Scenario
Scenario	You have been tasked with conducting PM on the 140 kW MG's.	
Activity Steps	Group Discussion: (25 minutes) *Instructor will facilitate a discussion pertaining to safety considerations.* • PPE (gloves, mask, etc.) • Lock-out (3 sources) ▪ Importance of insulation resistance checks • MG during insulation resistance checks Video with Discussion: (75 minutes) *Instructor will start and stop video and explain throughout.* • PMG ▪ Belt inspection ▪ Belt tensioning ▪ Belt replacement ▪ Consequences of PMG malfunction (AC power disrupted) Guided Instruction with PowerPoint augmentation: (100 minutes) • Commutator ▪ Scoring ▪ Arcing ▪ Resurfacing ▪ Brushes ▪ Percentage remaining ▪ Marking (percentage) ▪ Types of brushes ▪ Springs (seating) ▪ Proper seating Scenario: (50 minutes) *Instructor will provide explanation of insulation resistance checks troubleshooting.* • Insulation resistance checks ▪ Checking for proper grounding point ▪ Commutator		

LG 006.01.01 – 140kW MG	
	▪ Cleaning procedure ▪ Accessibility ▪ Brush holders ▪ Polishing ▪ Carbon tracking ▪ Vacuum ▪ Bearing ▪ Look-away technique
Instructor Resources	**References**
1) Safety considerations list 2) PMG video 3) Commutator PowerPoint 4) Scenario on insulation resistance checks troubleshooting 5) Computer 6) Screen	1) B1, B67, B68, B69, B70, B72, B73, B74, B75, B76, B108, B118
Note(s) to Instructor	
1) N/A	

EO 006.01 – 140KW MG
EC 006.01 – 140KW MG

EC 006.01 – 140kW MG		
Scope	**Location Conditions**	**Instructional Sequence**
EO 006.01	S100 Classroom	Day after LG 006.01.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Scenario	You have been tasked with conducting PM on the 140 kW MG's.	
Assessment Steps	**Input** – Students will watch video with correct procedure next to video with incorrect procedure. **Performance** – Students will have to identify differences between the videos by identifying mistakes. Video will be ~10 minutes and will play 3 times over course of EC. **Output** – The trainee must identify 70% of mistakes to pass this EC. **Feedback** – From instructor to student individually following completion of assessment.	
Instructor Resources		
1) Correct procedure video 2) Incorrect procedure video 3) Screen 4) Computer 5) PP&S		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	Nil.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 006.01 Maintain 140kW MG			
Item	**SAT**	**UNSAT**	**Remarks**
1. Video Errors identified			
PPE	☐	☐	**CRITICAL**
Lock out	☐	☐	
Belt tensioning	☐	☐	
Brushes (marking)	☐	☐	
Dropping springs	☐	☐	
Grounding point	☐	☐	
Discharge after IR reading	☐	☐	
Speed probe	☐	☐	
	☐	☐	

Score: /8

Trainee must achieve 5/7 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 006.02 – AC MOTORS AND STARTERS
LG 006.02.01 – AC MOTORS AND STARTERS

LG 006.02.01 – AC Motors and Starters		
Location Conditions		**Instructional Sequence**
S100 Classroom		After LG 006.01.01
Time	**Classification**	**Instructor**
250 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) PM 2) Faults 3) Safety precautions 4) IR 5) Calibrating electro-hydraulic (Sirco) control valve 6) Auto lubrication control system 7) VCP		1) Group Discussion 2) Scenarios 3) Think, Pair, Share 4) Guided Instruction 5) Video
Scenario	You have been tasked with conducting PM on the Stbd SW Pp Motor and Starters.	
Activity Steps	Group Discussion: (50 minutes) *Students will receive PM on AC motor and starters then read individually and discuss as a group. While this discussion takes place AC motor/starter is on screen.* • PM on AC motor • Components & normal voltage • Operational sequence of AC starter Scenarios: (75 minutes) *Instructor will give examples on common faults.* • AC motor/starter faults to include resistance and voltage readings and visible defects. ▪ Faulty contactor ▪ Faulty relay ▪ Faulty MPTU (Motor Protection Thermo Unit) card Think, Pair, Share: (50 minutes) *Students will write down what they believe are the personnel and equipment safety considerations individually. They will then discuss their answers in pairs. Each pair will then share their views with the group. Instructor will add anything missed and make corrections as required.* • Live troubleshooting ▪ Personnel safety requirements ▪ 2nd person with cane ▪ Gloves ▪ Equipment Safety Requirements ▪ Short circuiting • De-energized troubleshooting ▪ Personnel safety requirements	

LG 006.02.01 – AC Motors and Starters	
	▪ Isolate power ▪ Test for power ▪ Discharge filter ▪ Equipment Safety Requirements ▪ Removal of MPTU card ▪ Insulation resistance readings/meggering Guided Instruction with PowerPoint augmentation: (25 minutes) • Permissible IR reading ranges • Procedure if IR reading out of range • Causes of IR reading degradation Video with Discussion: (25 minutes) • Calibrating electro-hydraulic (Sirco) control valve Guided Instruction with PowerPoint augmentation: (25 minutes) • Auto lubrication control system ▪ Control box ▪ Troubleshooting ▪ Repairing • VCP (ventilation control panel) & associated equipment ▪ Duct heaters ▪ Ventilation fans ▪ Troubleshooting ▪ Repairing
Instructor Resources	**References**
1) DRMIS maintenance routine 2) Components and normal voltage handout 3) Scenarios with common AC motor and starters faults 4) PowerPoint on IR readings 5) Video of calibrating electro-hydraulic control valve 6) PowerPoint on auto lubrication control system and VCP 7) Computer 8) Screen 9) Projector 10) PP&S	1) B1, B67, B68, B69, B70, B72, B73, B74, B75, B76, B108, B118
Note(s) to Instructor	
1) N/A	

EO 006.02 – AC MOTORS AND STARTERS
EC 006.02 – AC MOTORS AND STARTERS

EC 006.02 – AC Motors and Starters		
Scope	**Location Conditions**	**Instructional Sequence**
EO 006.02.01	S100 Classroom	Day after LG 006.02.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Scenario	You have been tasked with conducting PM on the Port SW Pp Motor and Starters.	
Assessment Steps	**Input** – Students are expected to answer questions on PM on AC motor and AC starter. **Performance** – Students will answer written questions on personnel and equipment safety requirements on AC motor, identify minimum permissible IR readings, identify 70% of components on AC starter diagram, and list at least 3 PM checks for AC starter. **Output** – The trainee must achieve 4/5 SATs to pass this EC. **Feedback** – From instructor to student individually following completion of assessment.	
Instructor Resources		
1) Written test 2) AC starter diagram 3) PP&S		
Note(s) to Instructor		
1) N/A		
Assessment Limitations	Nil.	
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.	

PROTECTED B (when complete)

EC 006.02 AC MOTORS AND STARTERS			
Item	**SAT**	**UNSAT**	**Remarks**
1. Conduct PM on AC Motor			
Identify personnel safety requirements (minimum 3)	☐	☐	
Identify equipment safety requirements (minimum 2)	☐	☐	
Identify minimum permissible IR readings	☐	☐	
2. Conduct PM & Troubleshoot AC Starter			
Identify all components of AC starter (minimum 70% of components labelled)	☐	☐	
List PM checks for starter (minimum 3)	☐	☐	
	☐	☐	
	☐	☐	

Score: /5

Trainee must achieve 2/3 SATs on non-critical factors AND all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 006.03 – SFCS & TRUS
LG 006.03.01 – SFCS & TRUS

LG 006.03.01 – SFCS & TRUS		
Location Conditions		**Instructional Sequence**
S100 Classroom		After LG 006.02.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) SFC components 2) TRUs troubleshooting		1) Guided Instruction 2) Video
Scenario	You have been tasked with conducting PM and CM on the SFCs and TRUs.	
Activity Steps	Guided Instruction with PowerPoint Augmentation: (25 minutes) *Instructor will identify and discuss the internal components of the SFC with a picture open on PowerPoint.* • Troubleshooting SFCs • Identifying defects • Replacing components Video and Discussion: (25 minutes) *Instructor will introduce TRUs and show video of load balancing. After instructor will hold class discussion.* • Troubleshooting TRUs ▪ Load balancing TRUs ▪ Identifying defects ▪ Replacing components/unit ▪ Ordering parts • Troubleshooting SFCs ▪ Identifying defects ▪ Replacing components ▪ Ordering parts	
Instructor Resources		**References**
1) SFC PowerPoint 2) Computer 3) Screen 4) Video on TRUs load balancing		1) B1, B67, B68, B69, B70, B72, B73, B74, B75, B76, B108, B118
Note(s) to Instructor		
1) N/A		

PO 007 – MAINTAIN AUTO VOLTAGE & FREQUENCY REGULATORS

1. **Performance**: Maintain Auto Voltage & Frequency Regulators.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Operational AVFRs;
      (3) Electrical tools;
      (4) PPE; and
      (5) Test equipment.

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain Auto Voltage & Frequency Regulators by:

   a. Troubleshooting 140kW MG (Motor Generator) AVFR (Mk 201) (AUT0800, AUT2000, AUT3930, AUT2890, AT0080) by:

      (1) Identifying faults;
      (2) Isolating power to equipment;
      (3) Identifying cards; and
      (4) Changing cards.

   b. Troubleshooting slow speed MG AVFR (Mk 202) (AUT0800, AUT2000) by:

      (1) Identifying faults;
      (2) Isolating power to equipment;
      (3) Identifying cards; and
      (4) Changing cards.

   c. Troubleshooting diesel generator AVFR (Mk 203) (AUT0800, AUT2000) by:

      (1) Identifying faults;
      (2) Isolating power to equipment;
      (3) Identifying cards; and
      (4) Changing cards.

4. **Training Limitations**: Only available on submarine platform, unable to practice in trainer. Trainer in S-100 is inaccurate and doesn't provide useful training.

EO 007.01 – AVFRS
LG 007.01.01 – AVFRS

LG 007.01.01 – AVFRs		
Location Conditions		**Instructional Sequence**
S100 Classroom		After LG 006.03.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) Basic functionality 2) Adjustments & control 3) Features 4) Difference		1) Guided Instruction
Scenario	You have been tasked with conducting PM on the MK 201.	
Activity Steps	Guided Instruction with PowerPoint Augmentation: (50 minutes) *Instructor introduces the three AVFRs.* • Basic functionality • Adjustments & MCC control (Mk 203) • Common features • Differences	
Instructor Resources		**References**
1) PowerPoint on AVFRs 2) Computer 3) Screen		1) B1, B67, B68, B69, B70, B72, B73, B74, B75, B76, B108, B118
Note(s) to Instructor		
1) N/A		

PO 008 – MAINTAIN CSS & ALARM SYSTEMS

1. **Performance**: Maintain CSS & Alarm Systems.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Fire detection system;
      (3) CSS;
      (4) Alarm systems;
      (5) Electrical tools; and
      (6) Test equipment.

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside or at sea.

3. **Standard**: The Maintainer shall Maintain CSS & Alarm Systems by:

   a. Maintaining CSS by:

      (1) Operating surveillance system locally (YT0210); and
      (2) Operating surveillance system remotely (YT0220) by:
         (a) Reading feedback then:
            i. If LSU malfunctions:
               1) Resetting LSU;
               2) Repairing defects;
               3) Replacing parts on card rack; and
               4) Ordering parts as required.
            ii. If channel out of range repairing field devices.

   b. Inspecting MCC/SCC control by:

      (1) Maintaining navigation lighting controls by:
         (a) Inspecting mast flexible cables (KT0210).
      (2) Testing liquid detector module;
      (3) Testing alarms by:
         (a) Performing function test;
         (b) Troubleshooting faulty alarms by:
            i. Inspecting relay contacts (KT0240); and

ii. Repairing field devices.

(c) Testing alarm relay outputs.

(4) Testing emergency shutdown switches by:

(a) Checking functionality; and

(b) Repairing defects if required.

c. Maintaining control systems by:

(1) Maintaining FDS from motor room (IT0270, ANT1670) by:

(a) Testing FDS field devices by:

i. Performing function test;

ii. Repairing defective parts, or;

iii. Replacing defective parts; and

iv. Ordering parts as required.

(b) Testing FDS system (IT0270, ANT1670) by:

i. Performing FDS function test;

ii. Repairing defective parts, or;

iii. Replacing defective parts; and

iv. Ordering parts as required.

(2) Maintain control panels by:

(a) Troubleshooting diesel generator control panel (KT0660) by:

i. Repairing diesel generator control panel;

ii. Calibrating circuit cards;

iii. Adjusting circuit card connections;

iv. Inserting circuit card; and

v. Performing function test.

(b) Troubleshooting chilled water control panels (KT0700) by:

i. Repairing chilled water control panels (KT0340);

ii. Calibrating circuit cards;

iii. Adjusting circuit card connections;

iv. Inserting circuit card; and

v. Performing function test.

4. **Training Limitations**: Only have a single Teddington control panel in S-100, if malfunctions then must be conducted on submarine.

EO 008.01 – CSS
LG 008.01.01 – CSS

LG 008.01.01 – CSS		
Location Conditions		**Instructional Sequence**
S-100 CSS Trainer / S100 Classroom		After LG 007.01.01
Time	**Classification**	**Instructor**
400 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) Operating locally & remotely 2) Reading feedback 3) LSU malfunctions 4) Card rack 5) Field devices		1) Guided Instruction 2) Trainer
Scenario	You have been tasked with investigating a fault on the CSS.	
Activity Steps	Guided Instruction with PowerPoint Augmentation (50 minutes) *Instructor will introduce CSS using PowerPoint with pictures of system.* Trainer with Discussion: (350 minutes) *Instructor brings students to the CSS Trainer to explain and demonstrate CSS.* • Operating locally & remotely • Reading feedback • LSU malfunctions • Card rack • Field devices	
Instructor Resources		**References**
1) CSS Manual 2) Portable Operating Unit (POU) 3) CSS trainer 4) Computer 5) Screen		1) B1, B11, B12, B17, B18, B19, B20, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B54, B109, B119
Note(s) to Instructor		
1) If CSS Trainer unavailable will need Victoria Class Submarine.		

EO 008.02 – INSPECT MCC/SCC CONTROL
LG 008.02.01 – INSPECT MCC/SCC CONTROL

LG 008.02.01 – Inspect MCC/SCC Control		
Location Conditions		**Instructional Sequence**
S-100 CSS Trainer / S100 Classroom		After LG 008.01.01
Time	**Classification**	**Instructor**
250 minutes		AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) Navigational lighting controls 2) Generator liquid detector module 3) Alarm function test 4) Relay contacts 5) Field devices 6) Relay outputs 7) Emergency shutdown switches		1) Interactive Demonstration
Scenario	You have been tasked with conducting PM on the SCC.	

Activity Steps	Card Sort: (25 minutes)
	Instructor to hand out 3 stacks of cards, one stack with all lights, one stack with all proper names, and one stack with locations. Students will attempt to match the light to name to location. Instructor will provide DS Solution and review.
	Navigational lighting controls (7 or 8 lights)
	- Mast head steaming light
	- Stern running light
	- Port running light
	- Starboard running light
	- Forward and Aft anchor light
	- Anti-collision light (Grimes)
	- Not under command light
	
	Interactive Demonstration on Submarine Platform: (100 minutes)
	Instructor will take students to Submarine platform and show them the location and how to test the generator liquid detector module.
	
	Interactive Demonstration on Submarine Platform: (100 minutes)
	Instructor will take students to Submarine platform and open up doors on SCC to test alarms.
	- Testing alarms
	- Alarm function test
	- Troubleshoot phantom/false alarms
	- Install and remove IPBs
	- Field devices
	  o Limit switch
	  o Float floods
	  o Thermistors
	  o Level sensors	

LG 008.02.01 – Inspect MCC/SCC Control	
- Emergency shutdown switches	
Instructor Resources	**References**
1) List of lights, name, and location 2) 3 card sort stacks 3) Submarine platform	1) B1, B11, B12, B17, B18, B19, B20, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B54, B109, B119
Note(s) to Instructor	
1) If CSS Trainer unavailable will need Victoria Class Submarine.	

EO 008.03 – FDS
LG 008.03.01 – FDS

LG 008.03.01 – FDS		
Location Conditions		**Instructional Sequence**
S-100 FDS Trainer / S100 Classroom / Submarine Platform		After LG 008.02.01
Time	**Classification**	**Instructor**
50 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) FDS field devices 2) FDS system		1) Interactive Lecture
Scenario	You have been tasked with conducting PM and CM on the FDS.	
Activity Steps	Interactive Lecture (50 minutes) *Instructor brings student to the FDS Trainer and explain then demonstrate local operation.* • FDS field devices ▪ Performing function test ▪ Repairing defective parts, ▪ Replacing defective parts • FDS system ▪ Performing FDS function test ▪ Repairing defective parts, or ▪ Replacing defective parts	
Instructor Resources		**References**
1) Testing kit 2) Field devices 3) FDS trainer		1) B1, B11, B12, B17, B18, B19, B20, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B54, B109, B119
Note(s) to Instructor		
1) If FDS Trainer unavailable will need Victoria Class Submarine. 2) S-100 trainer needs updated field devices and testing kit		

EO 008.04 – MAINTAIN CONTROL PANEL
LG 008.04.01 – MAINTAIN CONTROL PANEL

LG 008.04.01 – Maintain Control Panel		
Location Conditions		**Instructional Sequence**
S100 Classroom/Submarine Platform		After LG 008.03.01
Time	**Classification**	**Instructor**
100 minutes	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) Troubleshooting diesel generator control panel 2) Troubleshooting chilled water control panels		1) EDI 2) Guided Discussion
Scenario	You have been tasked with conducting PM and CM on the diesel LCP.	
Activity Steps	Guided Instruction with PowerPoint augmentation: (25 minutes) • Troubleshooting diesel generator control panel - SIM test - Adjust set points - Calibrate cards - Function test Teddington Trainer: (50 minutes) • Troubleshooting diesel generator control panel - Calibrating circuit cards (DIL switches) - Adjusting circuit card connections - Inserting circuit card - Install card extender Scenario: (25 minutes) *You have an alarm tripping on the chilled water control panel.* Troubleshooting chilled water control panels - Repairing chilled water control panels - SIM test	
Instructor Resources		**References**
1) Teddington trainer 2) PowerPoint on diesel generator control panel 3) Computer 4) Screen 5) Scenario on troubleshooting chilled water control panels		1) B1, B11, B12, B17, B18, B19, B20, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B54, B109, B119
Note(s) to Instructor		
1) If CSS Trainer unavailable will need Victoria Class Submarine		

PO 009 – CONDUCT LONG & SHORT PRE-SEAS

1. **Performance**: Conduct Long & Short Pre-Seas.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Main battery & associated systems;
      (3) Electrical tools;
      (4) Cut out devices;
      (5) Safety equipment; and
      (6) Test equipment.

   b. Denied: Nil; and

   c. Environment:

      (1) Operational submarine; and
      (2) Alongside.

3. **Standard**: The Maintainer shall Conduct Long & Short Pre-Seas by:

   a. Completing long pre-seas (ZT0020) two weeks prior to sailing IAW VCSOPs to include, but not limited to:

      (1) Conducting system line ups;
      (2) Verifying electrical systems;
      (3) Verifying propulsion switchboard;
      (4) Verifying SCC systems;
      (5) Verifying engine room functionality;
      (6) Verifying MCC functionality;
      (7) Verifying VC-AS & OMC; and
      (8) Verifying surveillance systems.

   b. Completing short pre-seas (ZT0020) within 24 hours prior to sailing IAW VCSOPs to include, but not limited to:

      (1) Conducting system line ups;
      (2) Verifying electrical systems;
      (3) Verifying alarm systems;
      (4) Verifying propulsion switchboard;
      (5) Verifying SCC systems;
      (6) Verifying engine room functionality;

(7) Verifying MCC functionality; and
(8) Verifying VC-AS & OMC.

c. Reporting to Chief Electrician upon completion; and

d. Performing ship wide electrical function test by:
(1) Testing all systems to include:
(a) Navigation lighting;
(b) Propulsion switchboard to include:
i. Testing field supply converters (KT0510); and
ii. Testing EFRU.
(c) VC-AS.

4. **Training Limitations**: VCS is required for this PO.

EO 009.01 – LONG & SHORT PRE-SEAS
LG 009.01.01 – LONG & SHORT PRE-SEAS

LG 009.01.01 – Long & Short Pre-Seas		
Location Conditions		**Instructional Sequence**
S100 Classroom/Submarine Platform		After LG 008.04.01
Time	**Classification**	**Instructor**
500 min	Unclas	AIMM Qualified
Learning Objectives		**Instructional Method(s)**
1) System line ups 2) Electrical systems 3) Propulsion switchboard 4) SCC systems 5) Engine room functionality 6) MCC functionality 7) VC-AS & OMC 8) Surveillance systems 9) Alarm systems		1) Think-Pair-Share
Scenario	You have been tasked with conducting long pre-seas.	
Activity Steps	Think-Pair-Share: (50 minutes) *Students will write down the following individually. They will then discuss their answers in pairs. Each pair will then share their views with the group. Instructor will add anything missed and make corrections as required* • What they are • When you do them • What to do • Who to report to Trainer: (50 minutes) Verifying propulsion switchboard Platform: (400 minutes) Conducting system line ups Verifying electrical systems Verifying SCC systems, Verifying engine room functionality Verifying MCC functionality Verifying VC-AS & OMC Verifying surveillance systems	
Instructor Resources		**References**
1) Operational VCS		1) VCSOP Vol 1&2 2) VCSSO 3) Long & Short pre-sea checks
Note(s) to Instructor		
1) N/A		

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. All ECs are formative assessments that help the trainee and the instructors to recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

3. The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

4. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

5. Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

7. Continuous monitoring of trainee progress is required to provide the following:

8. Early warning of trainee difficulties;

9. Feedback on the effectiveness of training;

10. Information for the Training Review Board (TRB) (if required); and

11. Information to assist in the completion of the CF 377 Course Report.

12. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

13. PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

14. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

15. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

16. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

17. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

18. A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defence Subject Classification and Disposition System (DSCDS) Primary Number 4500.

ANNEX A – REFERENCES

This table lists all references used within this QSP.

#	Short Title	Long Title
A – CAF or other Official Canadian Government Publications		
B – Official Allied Force and Allied Government Publications		
B1		BRF 1966(1) GENERAL INFORMATION INDEX
B2		BRF 1966(10) PROPULSION: MECHANICAL SYS AND EQUIPTS - INDEX
B3		BRF 1966(10) 01 INTRO TO POWER GEN, PROPULSION & CONTROL
B4		BRF 1966(12) VENT AIR CON & SNOR IND SYS & EQUIPTS - INDEX
B5		BRF 1966(12) 01 SHIPS VENTILATION SYSTEM
B6		BRF 1966(12) 02 SNORT INDUCTION SYSTEM
B7		BRF 1966(12) 03 SNORT INDUCTION HULL VALVE
B8		BRF 1966(12) 04 SNORT INDUCTION FLAP VALVE
B9		BRF 1966(12) 05 VENTILATION SYSTEM BUTTERFLY VALVE
B10		BRF 1966(12) 06 AIR TREATMENT UNITS
B11		BRF 1966(19) 01 SHIPS DEPTH SYSTEM
B12		BRF 1966(19) 02 SHIPS DEPTH ELECTRONICS ENCLOSURE
B13		BRF 1966(19) 03 SHIPS DEPTH RETRANSMISSION UNIT
B14		BRF 1966(19) 04 SHIPS SPEED & DISTANCE SYSTEM
B15		BRF 1966(19) 05 LOG SPEED & DISTANCE UNIT
B16		BRF 1966(2) FIREFIGHTING SYSTEMS & EQUIPMENTS – INDEX
B17		BRF 1966(2) 01 FIRE PREVENTION & DETECTION SYSTEM
B18		BRF 1966(2) 02 FIRE-DETECTION RACK AND FIRE DETECTORS
B19		BRF 1966(2) 03 FIRE-DETECTION ALARM & ALARMFLASHER MODULES
B20		BRF 1966(2) 04 FIREFIGHTING SYSTEMS
B21		BRF 1966(2) 05 FIREFIGHTING SYSTEMS HULL VALVE
B22		BRF 1966(20) SURVEILLANCE SYSTEM & EQUIPMENTS - INDEX

B23		BRF 1966(20) 01 SURVEILLANCE SYSTEM CONFIGURATION
B24		BRF 1966(20) 02 CENTRAL PROCESSOR UNIT
B25		BRF 1966(20) 03 LOCAL SCANNER UNITS
B26		BRF 1966(20) 04 SURVEILLANCE SYSTEM PLASMA DISPLAYS
B27		BRF 1966(20) 05 SURVEILLANCE SYSTEM KEYBOARDS
B29		BRF 1966(20) 06 SURVEILLANCE SYSTEM PRINTER
B30		BRF 1966(20) 07 SURVEILLANCE SYSTEM LSU STATUS PANELS
B31		BRF 1966(21) CONTROL CONSOLES & ALARM SYSTEMS – INDEX
B32		BRF 1966(21) 01 SUBMARINE CONTROL CONSOLE
B33		BRF 1966(21) 02 ONE MAN CONTROL CONSOLE
B34		BRF 1966(21) 04 MACHINERY CONTROL CONSOLE
B35		BRF 1966(21) 05 COMPARTMENT FLOOD ALARMS
B36		BRF 1966(21) 06 GRADE A & B ALARMS
B37		BRF 1966(22) RUDD & HYDPLNS: STRUC CONT SYS & EQUPTS – INDEX
B38		BRF 1966(22) 03 RUDDER & HYDROPLANES: CONTROL SYSTEM
B39		BRF 1966(22) 04 RUDDER & HYDROPLANES: POWER PLANT SYSTEM
B40		BRF 1966(22) 12 AUTOPILOT & RUDDER & PLANES ELECTRONIC ENCL
B41		BRF 1966(22) 13 OUTER LOOP MONITORING SYSTEM
B42		BRF 1966(23) MASTS & ASSOCIATED EQUIPMENTS – INDEX
B43		BRF 1966(23) 06 MASTS CONTROL & INDICATION CIRCUITS
B44		BRF 1966(4) VPDC System and Equipments -Index
B45		BRF 1966(4) 01 VPDC Distribution and Shore Supplies
B46		BRF 1966(4) 02 VPDC Earth Monitoring Equipment
B47		BRF 1966(4) 03 Battery Switchboards
B48		BRF 1966(4) 05 Main DC Circuit Breakers
B49		BRF 1966(4) 06 DIESEL-DRIVEN GENERATOR
B50		BRF 1966(4) 07 GENERATOR AIR COOLER
B51		BRF 1966(4) 08 DG AIR CONTROL PANEL – MK 203 CP
B52		BRF 1966(4) 09 DG SWITCHBOARDS
B53		BRF 1966(4) 10 VPDC LOAD CENTRES
B54		BRF 1966(5) MAIN BATTERY SYSTEMS AND EQUIPMENT - INDEX

B55		BRF 1966(5) 01 HYDROGEN MONITORING
B56		BRF 1966(5) 02 BATTERY VENT & HYDROGEN CLEARING SYSTEMS
B57		BRF 1966(5) 03 BATTERY VENT: ISOLATION SELECTORS & ACTUATORS
B58		BRF 1966(5) 04 BATTERY COOLING AND TOP-UP SYSTEM
B59		BRF 1966(5) 05 BATTERY AGITATION SYSTEM
B60		BRF 1966 (5) 06 BATTERY COOLANT PUMP
B61		BRF 1966 (5) 07 BATTERY EXHAUST FAN MOTOR
B62		BRF 1966 (5) 08 BATTERY EXHAUST FAN MOTOR STARTER
B63		BRF 1966 (5) 10 BATTERY COOLANT PUMP
B64		BRF 1966 (6) 24V DC SYSTEMS & EQUIPMENTS - INDEX
B65		BRF 1966 (6) 01 24V DC SYSTEM
B66		BRF 1966 (6) 02 TRANSFORMER RECTIFIER UNIT (24V DC)
B67		BRF 1966 (6) 03 24V DC DISTRIBUTION PANELS
B68		BRF 1966 (7) AC SYSTEMS & EQUIPMENTS – INDEX
B69		BRF 1966 (7) 01 440V 60HZ SYSTEM
B70		BRF 1966 (7) 02 200/115V 400HZ SYSTEM
B71		BRF 1966 (7) 04 140KW MOTOR GENERATOR
B72		BRF 1966 (7) 05 MG AIR COOLER
B73		BRF 1966 (7) 06 140KW CONTROL PANEL MK201 CP
B74		BRF 1966 (7) 07 140KW MG STARTER
B75		BRF 1966 (7) 08 AC SWITCHBOARDS
B76		BRF 1966 (7) 09 AC LOAD CENTRES
B77		BRF 1966 (7) 10 AC FUSE PANELS
B78		BRF 1966 (8) 02 DG SETS: INSTALLATION
B79		BRF 1966 (8) 03 DISEL ENG: AUTO CONTROL& PROTECTION SYSTEM
B80		BRF 1966 (9) PROPULSION: ELECTRICAL SYSTEMS & EQUIPMENT INDEX
B81		BRF 1966 (9) 01 INTRO TO POWER GEN, PROPULSION & CONTROL
B82		BRF 1966 (9) 02 PROPULSION SYSTEM
B83		BRF 1966 (9) 03 PROPULSION CONTROL SYSTEM
B84		BRF 1966 (9) 05 PROPULSION MOTOR
B85		BRF 1966 (9) 06 PROPULSION MOTOR BLOWER & STARTER
B86		BRF 1966 (9) 07 PROPULSION MOTOR AIR COOLER
B87		BRF 1966 (9) 08 SLOW-SPEED PROPULSION MOTOR GENERATOR Version [1.0]: Sub-Version [00] 3

B88		BRF 1966 (9) 09 SLOW-SPEED PROPULSION MG STARTER
B89		BRF 1966 (9) 10 SLOW-SPEED PROPULSION MG CONT PANEL – MK202 CP
B90		BRF 1966 (9) 11 PROPULSION: FIELD-SUPPLY CONVERTER
B91		BRF 1966 (9) 12 PROPULSION SWITCHBOARD
B92		BRF 1966 (9) 16 SHAFT-SPEED INDICATION SYSTEM
B93		BRF 1966 (9) 19 PROPELLER ORDER EQUIPMENT DIGITAL DISPLAY
B94		BRF 1966 (9) 20 PROPELLER ORDER EQUIPMENT OPTO-ISOLATOR
B95		BRF 1966 (9) 23 PROP MOTOR MAIN BEARING LUBRICATION SYSTEMS
B96		BRF 1966 (9) 24 OVERSPEED LIMIT UNIT
B97		BRF 1966 (9) 25 EXCITATION FAILURE RELAY UNIT
B98		BRF 2075 (K) SUBMARINE MAIN BATTERY VICTORIA CLASS
B99		BRF 4500 (526) MK 38 SFC
B100		BRF 4500 (529) MK 38 SFC
B101		BRF 4540(101) WHIPPENDALE TYPE MCA2 & MCA3 CONTRACTORS
B102		BRF 4540(102) WHIPPENDALE TYPE MCA5 CONTRACTORS
B103		BRF 4540(103) WHIPPENDALE TYPE MCA13, 20, 25 & MCA 25s CONTACTORS
B104		BRF 4540(104) ALLEN WEST TYPE SCF CONTACTOR
B105		BRF 4540(109) EAC TYPE A20 AC CONTACTOR
B106		BRF 4540(112) WATFORD CONTACTORS TYPES UCA 1N, UCA 7N, UCA 20N
B107		BRF 4540(301) WATFORD OVERCURRENT RELAYS
B108		BRF 4540(503) THERMAL PROTECTION UNIT MK4
B109		BRF 4540(604) AC STARTERS
B110		BRF4569 PRESSURE & DIFFERENTIAL PRESSURE TRANSDUCERS & ASSOCIATED EQUIPMENT
B111		BRF6522(702) INVERTER LIGHTING & PSU’s
B112		BRF 6528(207) REGULATOR EUROPA-400 SERIES GOVERNOR
B113		BRF 1966 (13) 02 CHILLED WATER PLANT SYSTEM AND COMPONENTS
B114		BRF 1966 (13) 03 CHILLED WATER PLANT CONTROL AND INDICATION Version [1.0]: Sub-Version [00] 4
B115		BR2075(KX) SUB MAIN BATTERIES TYPE 8800T DOCKSIDE & REFIT PROCEDURES

B116		BR2120 INTERFERENCE SUPPRESSION
B117		BR3623(1) 01 SHIP SYSTEM - HMS UPHOLDER
B118		BR 3623(2) 01 PROPULSION SYSTEM - HMS UPHOLDER
B119		BR3623(4) 01 ELECTRICAL GENR & DISTR – HMS UPHOLDER
B120		BR4580(018) SURVEILLANCE SYSTEM
B121		BR6500 DC CIRCUIT BREAKERS
C – Commercial Publications		
D – Other Publications		
D1		ENERSYS TECHNICAL OPERATIONS MANUAL FOR VICTORIA CLASS SUBMARINE MAIN BATTERY TYPE 37KR177T
D2		VICTORIA CLASS AUTOPILOT SYSTEM (VCAS) OPERATIONS MANUAL
D3		VICTORIA CLASS AUTOPILOT SYSTEM (VCAS) MAINTENANCE MANUAL
D4		VICTORIA CLASS AUTOPILOT SYSTEM (VCAS) OMC INTERCONNECT DIAGRAM
D5		VICTORIA CLASS AUTOPILOT SYSTEM (VCAS) HARDWARE DESIGN DOCUMENT (HDD)
D6		VICTORIA CLASS AUTOPILOT SYSTEM (VCAS)STICK WHEEL (SW)

ANNEX B – TASK LIST

This table provides an overview of MTL tasks WRT if/where they are trained within this QSP. No train substantiations are located in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
AFT0400	Report maintenance activities within defence resource management system (DRMIS)	T	003
AKT0800	Inspect visually, carbon brushes	T	003
AKT1080	Investigate system maintenance logbooks	T	001
AKT1090	Investigate system operation logbooks	T	001
AKT1250	Manage preventative maintenance routines	T	003
AKT2600	Supervise engineering system maintenance	T	001
ANT1670	Troubleshoot fire detection system	T	008
AT0070	Conduct resistance test	T	001
AT0080	Conduct rotational speed test	T	007
AUT0800	Inspect automatic voltage regulator	T	007
AUT1450	Inspect solenoid	T	002
AUT1460	Inspect speed sensors	T	002
AUT1470	Inspect static frequency converter	T	006
AUT1960	Repair battery	T	001
AUT1970	Repair battery assemblies	T	001
AUT2000	Repair automatic voltage regulator	T	007
AUT2850	Repair transformer-rectifiers	T	006
AUT3170	Test DC motor	T	006
AUT3930	Test variable speed drive controller	T	007
AUT4090	Update battery log	T	001
BAT0090	Inspect electric propulsion systems	T	004
BAT0160	Investigate operation of auto pilot system	T	005
BAT0380	Troubleshoot auto pilot system	T	005
BT0220	Troubleshoot electric propulsion systems	T	004
BT0260	Troubleshoot propulsion motor	T	004
IT0270	Investigate operation of fire detection system	T	008
KT0140	Inspect battery cooling header pipe (sub)	T	001
KT0150	Inspect high voltage tube (sub)	T	002
KT0160	Inspect battery cooling fish plates (sub)	T	001
KT0180	Inspect CO2 absorption unit DC regulator (sub)	T	003
KT0190	Inspect generator rectifier (subs)	T	002
KT0200	Inspect hydrogen eliminators (sub)	T	001
KT0210	Inspect mast flexible cable (sub)	T	008
KT0220	Inspect ram servo linkages (sub)	T	005
KT0240	Inspect relay contacts	T	008
KT0340	Repair chilled water system (subs)	T	008

KT0360	Repair generator rectifier (subs)	T	002
KT0500	Test DC starter (sub)	T	003
KT0510	Test field supply converters (sub)	T	004
KT0520	Test generator rectifier (subs)	T	002
KT0540	Test port and starboard computers (sub)	T	005
KT0550	Test position indicators (sub)	T	005
KT0560	Troubleshoot Auto lubrication system (sub)	T	006
KT0620	Troubleshoot hydraulic plant (subs)	T	006
KT0660	Troubleshoot main diesel engine start control panel (Sub)	T	008
KT0680	Troubleshoot ship's ventilation systems (subs)	T	006
KT0690	Troubleshoot submarine battery ventilation system	T	001
KT0700	Troubleshoot submarine chilled water system	T	008
KT0710	Troubleshoot submarine CO2 absorption unit	T	003
YT0020	Inspect visually, submarine battery ventilation system	T	001
YT0210	Operate surveillance system locally (sub)	T	008
YT0220	Operate surveillance system remotely (Sub)	T	008
ZT0020	Conduct engineering pre-sailing checks	T	009

ANNEX C – RECORD OF DECISIONS

The following summary provides insight to the decisions and reasoning resulting in this QSP.

Dates: 26 Feb to 13 Mar 20

Location: Building S-15 – 203, NTDC(A), CFB Halifax, Halifax, NS

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Guthrie	Neal	SLt		Subs Div TDO
Morris	Kristen	Lt		TDO OJT
Etheridge	Kevin	LS		Mar Tech OJE
Coady	Adam	PO1		Senior Instructor NFS(A)
Mohr	Geoffrey	PO1		Senior Electrician
Staal	Jules	PO1		Chief Electrician
Dixon	Jake	LS		Electrical Technician
Briere	Erik			MCT/SCT OSR

Working Group Objective:

1. The purpose of convening this working group was to…

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation

Discussions / Decisions:

2.

Wednesday, 26 Feb 20 – Day 1 of 7 - Details	Decisions / Remarks
Intro and Round Table • A round table of introductions was conducted. • Lt Morris provided an intro and detailed brief to the WG. • Availability of SMEs discussed, possible COAs identified for AIMM QSP completion. Decided to complete a split board based on submariner electrical requirements for putting the submarine back into the water.	
Doc Review • SMEs reviewed the previous AIMM QSP.	

• Considerations pertaining to Teddington training, availability of equipment (for example the diesel generator, speed probes, and fresh water heaters) discussed briefly. • Some discussion pertaining to DG in S-15 basement, it's a great training aid for this course and the only trainer available for this purpose. The amount of time the DG is used and the amount of space it takes up precludes it from being a viable trainer. If space is available, it would be ideal to save the trainer. • Some documents still have old MOSID and haven't been updated to Mar Tech. • The vision for the OJPR is similar to old 5's package, it won't require drawings and ideally will include the entire maintenance routine.	
TPA • Reviewed TPA written at a previous attempt to run AIMM QSP WG. • Additional criteria added and TPA placed in Annex L. • The main difference between what has been done for this course traditionally and what is currently being done is that personnel are taking the course at LS rank rather than OS rank.	
Directed Training • The board discussed moving to a directed training on board the subs. This would involve the instructor teaching this course to conduct lessons and parts thereof on one of the subs. They discussed potential unavailability of subs for this purpose and deemed it was a negligible risk. The course load is small and the course only runs once every year or two. The dates for AIMM can be changed to allow time on the sub for training purposes.	
Train/No Train • T/NT conducted with MTL for Mar Tech using only sections K and Y. Many tasks deemed to be NT and details are found in Annex B – Task List. • Some tasks the WG could not agree or imagine what the MTL was inferring, details in Annex B. • All tasks in OSS deemed train, the WG agreed that these tasks are much broader than those in the MTL.	
Scalar • SMEs replicated the scalar by writing down all the tasks from both OSS and MTL. • Many additional tasks added to amplify. (Scalars not included in this document as they were changed significantly after being written in PO format)	
Thursday, 27 Feb 20 – Day 2 of 7 - Details	**Decisions / Remarks**
Scalar	

• Scalars developed based on a systems view due to the nature of the job. Tasks can be conducted in a specific sequence, however, the sequence is largely dependent on the issue the maintainer is attempting to rectify. For readability and comprehension it was decided the systems breakdown makes the most sense. • Two additional scalar groups emerged outside the previous pattern: long pre-seas and documentation. Some documentation was added relevant groups (i.e. battery logs). DRMIS training is provided prior to this course and will therefore not be covered again. Long pre-seas developed into PO 009. • The language of some tasks was discussion, particularly pertaining to troubleshooting versus testing. Troubleshooting is often conducted at the PO2 level or above, however, it is system dependant. There is a mix of troubleshooting and testing within the scalar and some tasks were updated to be more accurate to the LS job. • The scalar was finalized into 9 main groups which were sorted into a logical order and labelled PO 1 through 9.	
Directed Training • Portions of POs for directed training include Teddingtons, diesel axillaries, and pyros. More will be added as development progresses.	
Equipment • The surveillance system will most likely be gone by the time the next AIMM serial runs due to an upgrade on the subs. This trainer won't be included in this QSP. • There is currently a lack of trainers for cell cut-outs. The group requested an LSR be submitted to 3D print extra bits for cell cut-outs, a cell top, and materials within a full cut-out kit for students to practice.	1. LSR to be submitted.
OJPR & Assessments • Discussed implications of placing PCs on OJPR including transferring training onto units and the importance of standards. The WG stressed that the training equipment is too inaccurate compared to on the subs and it wouldn't be prudent to qualify a member AIMM without completing tasks onboard. The WG added that many students learn hands-on and OJPR would be the ideal testing environment.	1.
Friday, 28 Feb 20 – Day 3 of 7 - Details	
Assessments • Discussion over testing PCs in class. Many of the trainers don't provide realistic trainer and on the job differs significantly. Some POs will be tested on an OJPR package. • CM should be used for scenario based testing. • VC-AS common failures should be used for testing.	

Equipment • Language in QSP discussed, WG decided to use the generic name for equipment with current brand name in brackets for continuity.	
• WG broke for minimum 1 week to allow SMEs time to complete time-sensitive work onboard sub.	
Tuesday, 10 Mar 20 – Day 4 of 7 - Details	
Trainer • Discussed possibilities on what to do with DG trainer. Possibility to place into W-7, called CPO2 Bruce Lilly who advised the building is slated for demolition. Lt(N) Reginald Simmons may be able to aid with relocation of trainer. PO1 Adam Coady mentioned a report about DG trainer underutilization which has led to its current perilous situation. Lt Kristen Morris to track down report on usage. • The Ops Cert course, diesel maintainer for Stokers, and the AIMM course all use the DG trainer.	1. Lt Kristen Morris to find DG trainer report.
LSR • WG requested a video of the precipitator with the probe to be made. • WG requested a video of the DC starter with numbering at each step to be used as a confirmation activity.	1. Lt Kristen Morris to submit 2 LSRs.
OJPR • PO 002 OJPR “Conduct breakdown readings” was removed as it is only done once every 18 months and students wouldn’t be able to complete their packages in a timely manner.	
Teaching Content • CO2 absorption unit is very simple. Students will need to have the knowledge of what to do but testing is unnecessary as it’s easily done on the job.	
Wednesday, 11 Mar 20 – Day 5 of 7 - Details	
Skills • Incorporation of skills into the QSP was discussed. Many of the skills involve using tools. These skills would be very difficult to simulate on training as the difficulty lies in maneuvering in tight spaces and being restricted. Some of these can be simulated on trainers but it won’t aid greatly with skill development as it doesn’t reflect the job conditions.	1.
Teaching Content • Function test for 006.01.01 will be covered in the OJPR • ACU function test was moved to PO 009. • 140kW MG function test moved to PO 009.	
Thursday, 12 Mar 20 – Day 6 of 7 - Details	
•	1.
•	1.
Friday, 13 Mar 20 – Day 7 of 7 - Details	

Review of QSP • Revisions were made to QSP based on SME feedback and input. • WB approved current draft QSP. • TDO to finalize the QSP. • WG broke for minimum 3 weeks due to time sensitive Submarine maintenance and COVID-19.	1. Lt Kristen Morris to revise QSP. 2. SLt Neal Guthrie to revise QSP. 3. PO1 Adam Coady to finalize and submit QSP.

Battery maintenance on platform (sub)
AVRs on boat

Follow Up Action:

Item	Action	OPI	Expected Date of Completion

Course History:

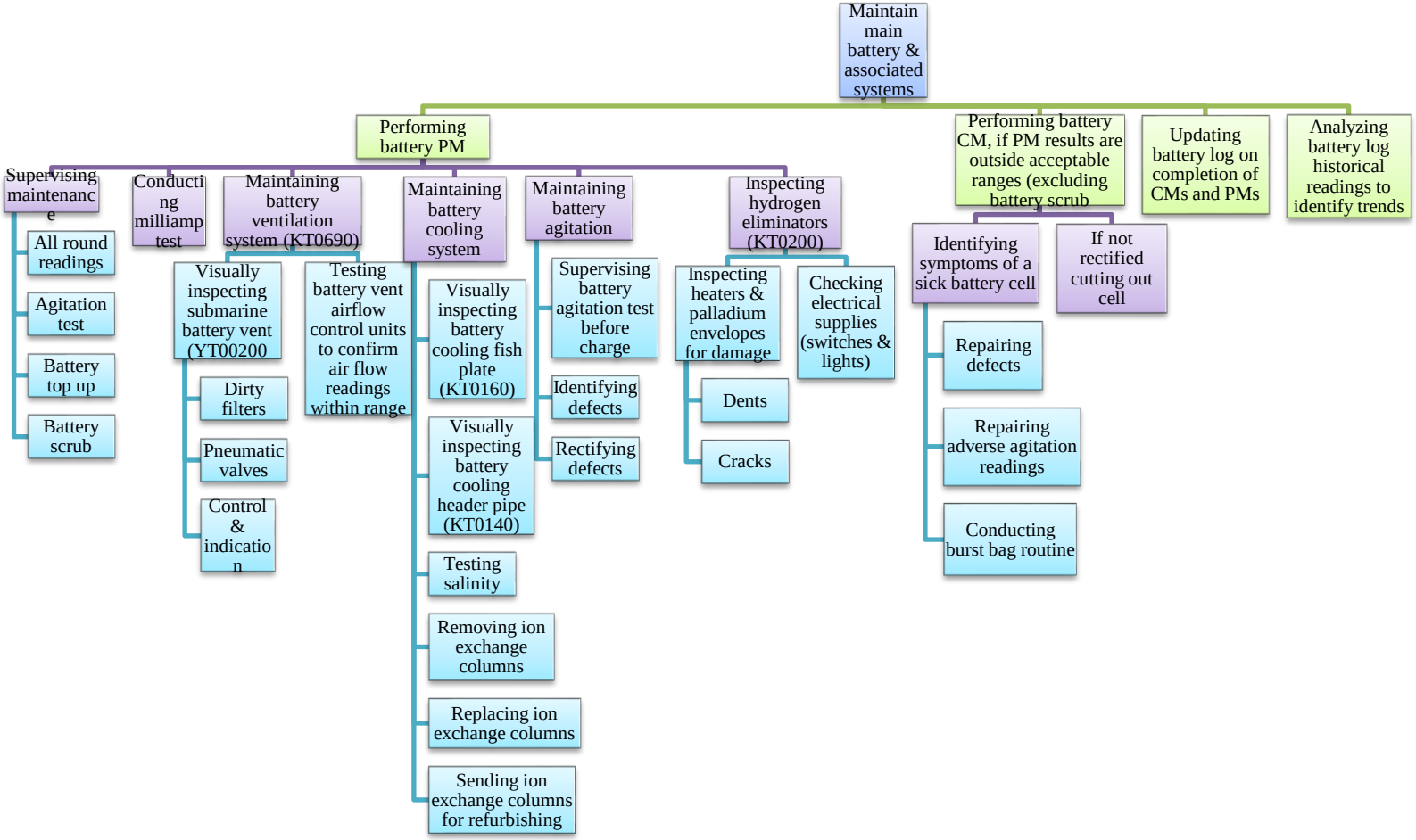
3. Include previous Records of Decisions here to give context to the evolution of the course.

22 Apr 2022, CPO2 Sutton Mar Tech Training Manager, added MITE Codes and removed highlighting of NQUAL. NQUAL being updated by DPGR to accept new MITE Codes.

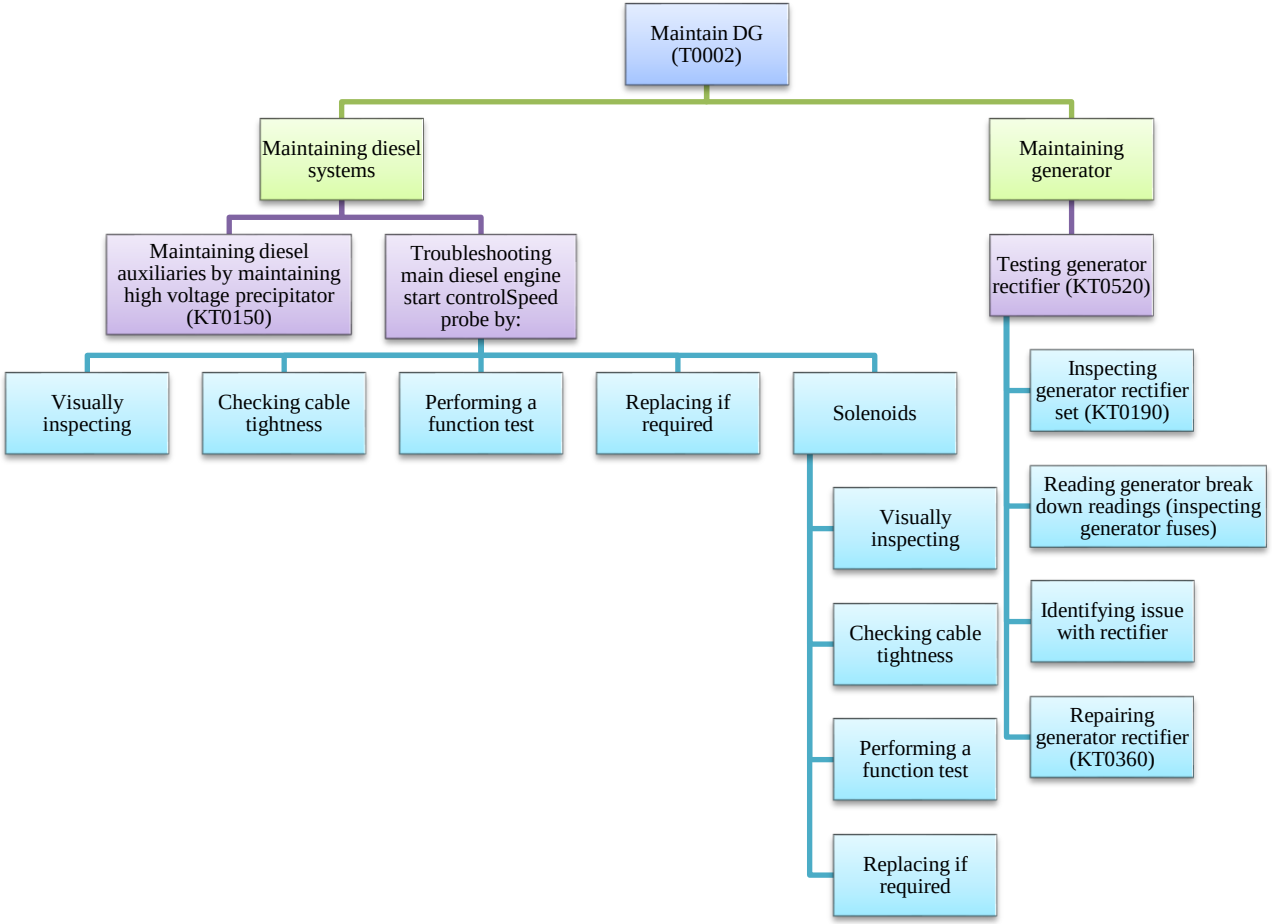
ANNEX D – TASK ANALYSIS DIAGRAM

This diagram represents the recreation or original task analysis (scalar) used to define the job.

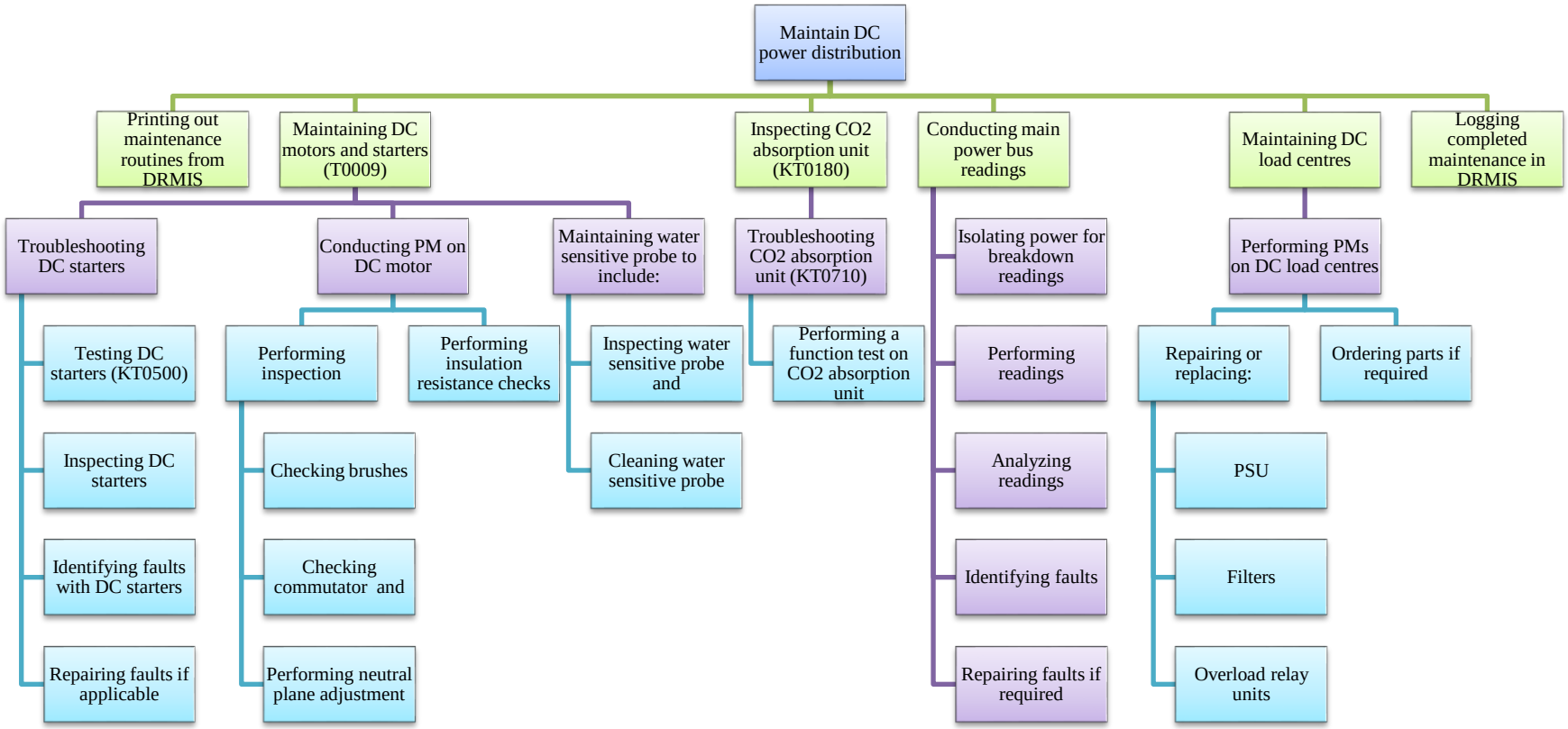
PO 001- Maintain Main Battery & Associated Systems



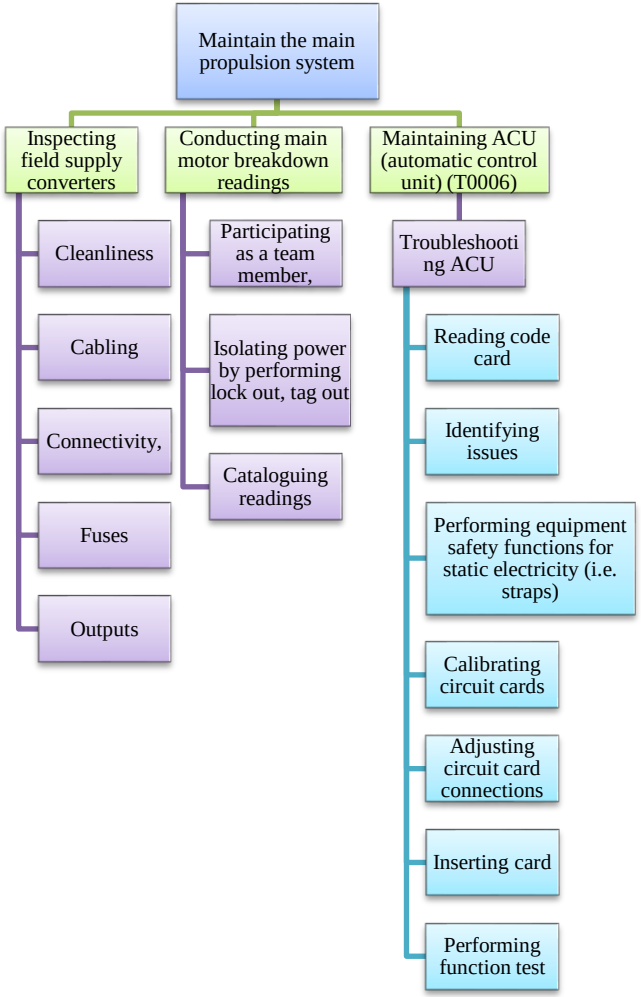
PO 002 – Maintain Diesel Generator



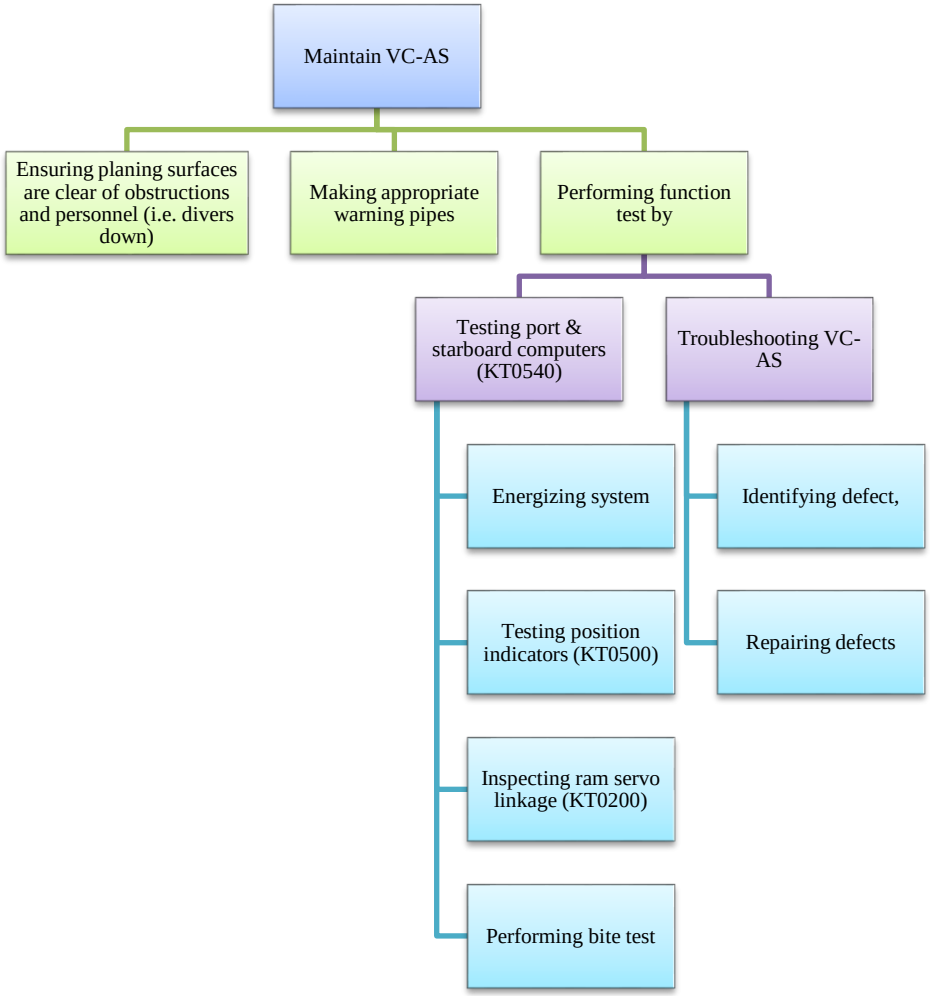
PO 003 – Maintain DC Power Distribution



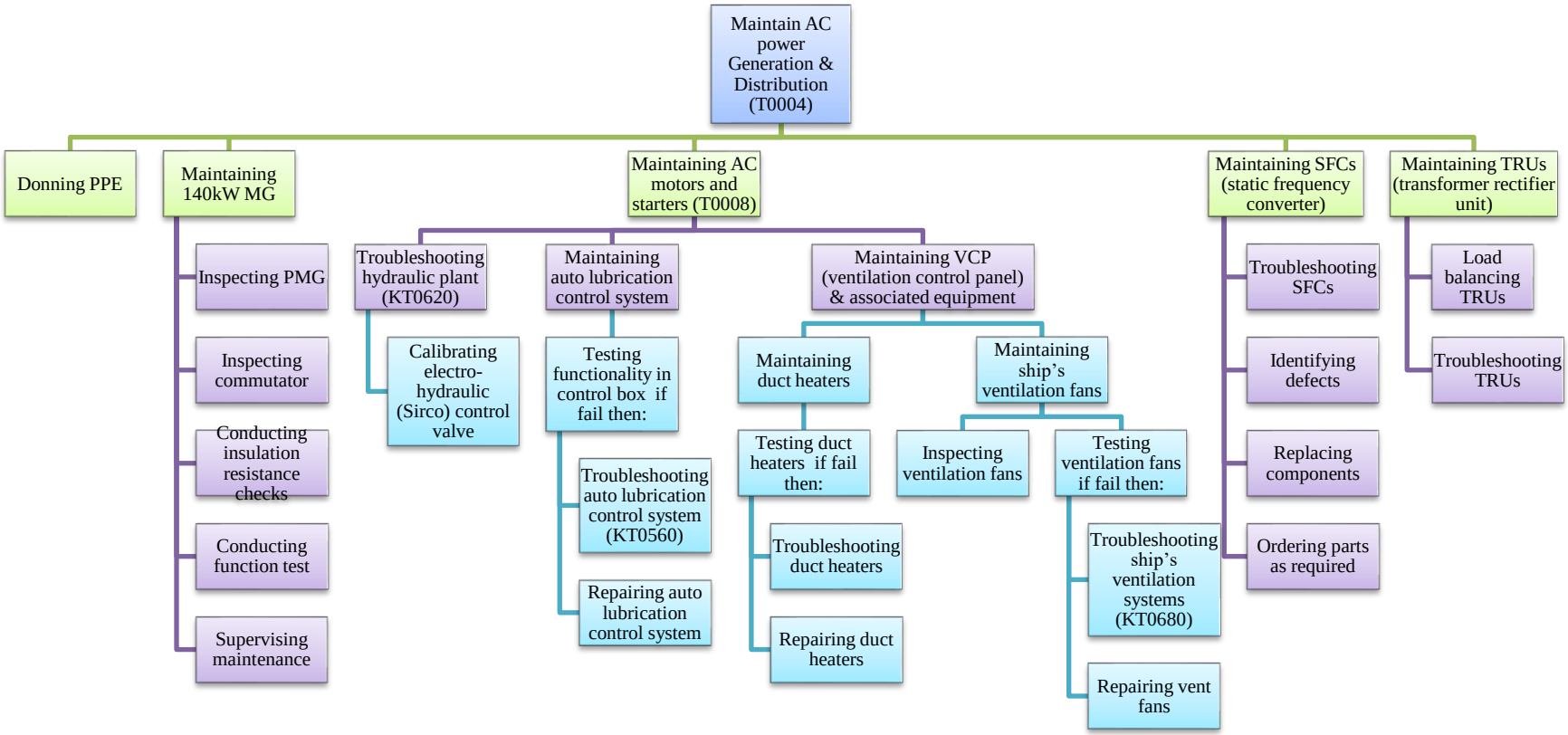
PO 004 – Maintain Main Propulsion System



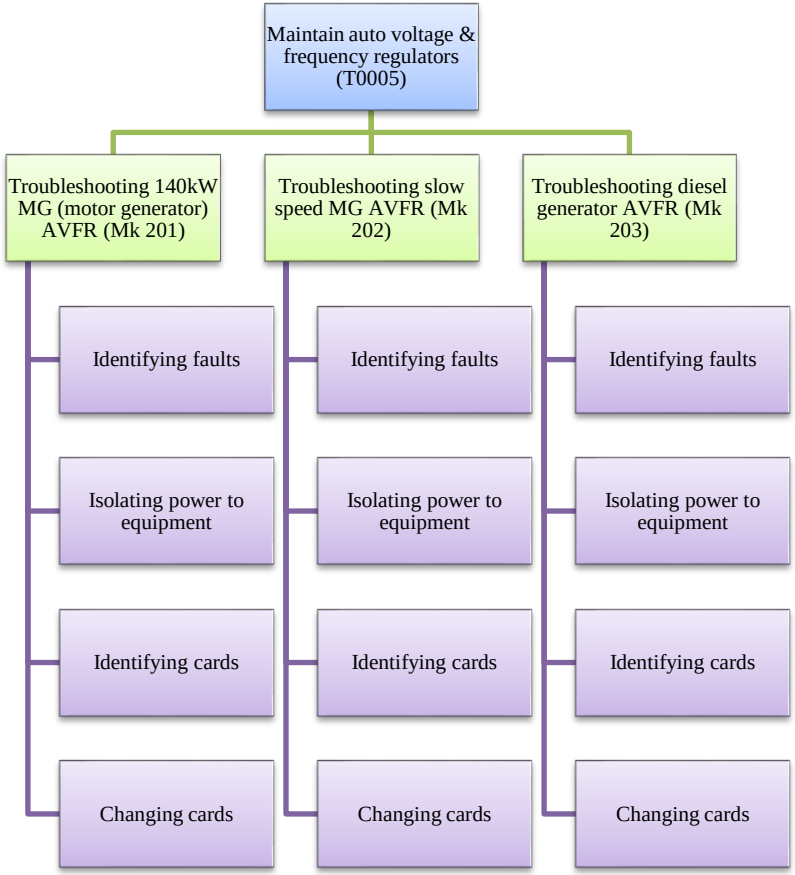
PO 005 – Maintain VC-AS



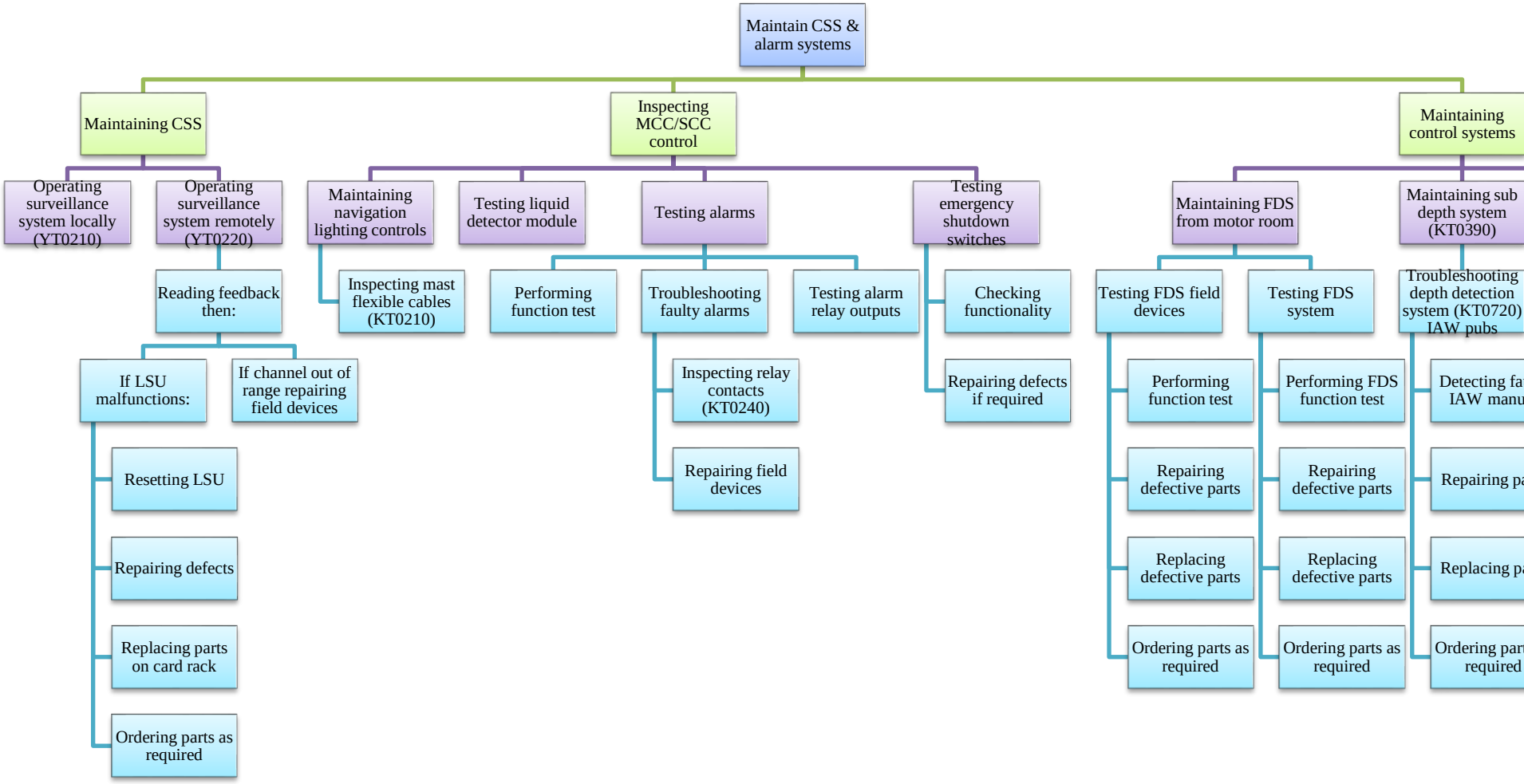
PO 006 – Maintain AC Power Generation & Distribution



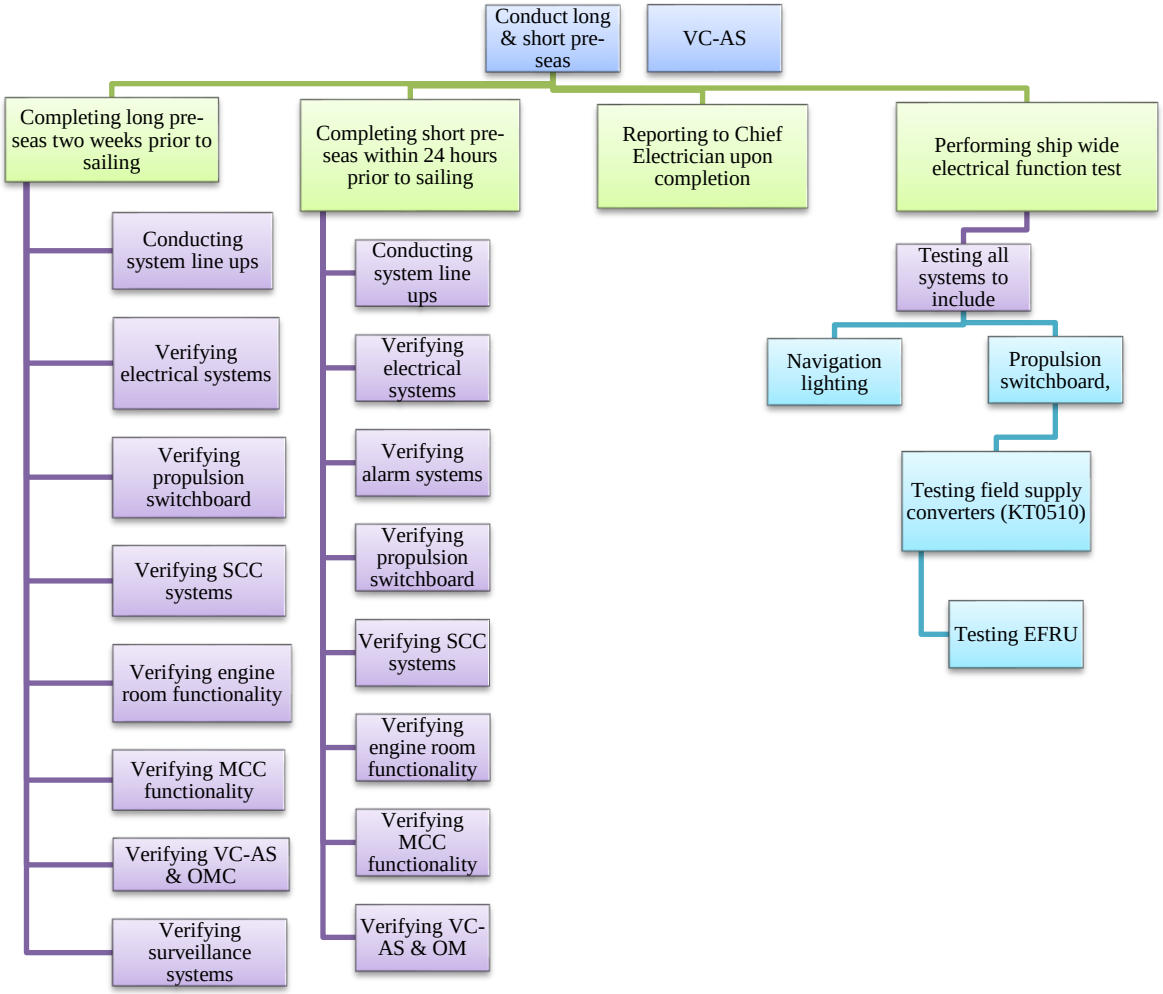
PO 007 – Maintain Auto Voltage & Frequency Regulators



PO 008 – Maintain CSS & Alarm Systems



PO 009 – Conduct Long & Short Pre-Seas



ANNEX E – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates. Personnel costs are IAW the Cost Factors Manual for the current fiscal year for all instructors identified in Chapter 1 - Required Course Staffing.

TD Costs	**# Trainees + Instructors /**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
Airfare (Return)	8	$1,000.00	n/a	$8,000	$8000	$40,000.00
R&Q	8	$36.00	23	$6,624	$6,624	$33,120.00
Incidentals	8	$17.50	23	$3,220	$3,220	$16,100.00
Subtotal				**$17,844**	**$17,844**	**$89,220.00**
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	6	$50.00	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.000	23			
Computer requirements	6	$1,000.00	1	$16,000	$0	$16,000.00
IT Equip (Smart board, etc.)	1	$6,000.00	1	$6,000	$0	$6,000.00
DWAN connection (Setup)	1	$5,000.00	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	6	$1.00	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	6	$30.00	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
Personnel Costs	**Quantity**	**Cost/day**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PO1	#	$###.##	###	$8,920	$8,920	$44,600.00
Civilian Instructor						
Subtotal				**$8,920**	**$8,920**	**$44,600.00**
Total Estimated Costs				**$55,029**	**$28,029**	**$167,145.00**

ANNEX F – ASSESSMENT PLAN

This table summarizes assessment (EC & PC) details contained within this QSP.

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01.01	To assess trainee's knowledge of Submarine battery preventative maintenance procedures.	Written	The trainee must achieve 10/14 SATs IAW EC 001.01 Checklist
EC 001.02	EO 001.02.01	To assess trainee's knowledge of Submarine battery corrective maintenance procedures.	Practical	Trainee must achieve 4/5 SATs on non-critical factors AND all critical factors IAW EC 001.02 Checklist
EC 002.01	EO 002.01	To assess trainee's knowledge of Submarine Diesel Generator Auxiliaries, solenoids, and speed probe.	Practical	Trainee must achieve 7/9 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 002.01 Checklist
EC 002.02	EO 002.02.01	To assess trainee's ability to maintain the Submarine Diesel Generator.	Practical	Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 002.02 Checklist
EC 003.01	EO 003.01.01	To assess trainee's knowledge of DC motors and starters maintenance procedures.	Written	Trainee must achieve 4/6 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 003.01 Checklist
EC 003.02	EO 003.02.01	To assess trainee's ability to take Main Power Bus readings.	Practical	Trainee must achieve 1/1 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 003.02 Checklist
EC 003.03	EO 003.03.01	To assess trainee's ability to maintain the submarine DC Load Centers.	Written	Trainee must achieve 1/1 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 003.03 Checklist

EC 004.03	EO 004.01.01 & EO 004.03.01	To assess trainee's knowledge of field supply converters and troubleshooting the ACU.	Practical	The trainee must achieve 4/6 SATs to pass this EC IAW EC 004.03 Checklist
EC 005.01	EO 005.01	To assess trainee's knowledge of Submarine VC-AS.	Practical	Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 005.01 Checklist
PC 005	EO 005.01	To assess trainee's ability to maintain the Submarine VC-AS.	Practical	Trainee must achieve 5/6 SATs on non-critical factors AND all critical factors to pass this PC IAW PC 005 Checklist
EC 006.01	EO 006.01.01	To assess trainee's ability to maintain the Submarine's 140kW MG.	Practical	Trainee must achieve 5/7 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 006.01 Checklist
EC 006.02	EO 006.02.01	To assess trainee's knowledge of AC Motors and Starters.	Written	Trainee must achieve 2/3 SATs on non-critical factors AND all critical factors to pass this EC IAW EC 006.02 Checklist

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

If a training methodology is determined to be ineffective and/or inefficient, a Learning Support Request (LSR) may be submitted to correct the deficiency. Technology Enabled Learning (TEL) development opportunities are assessed IAW NPTORD 4500-84 following submission.

Title	Requirement Description	Lesson #	LSR #
DC Motors & Starters	VR for AC/DC starters	LG 003.01.01	
Maintain Battery	Cell cut-out device & Cell top	LG 001.02.01	

ANNEX H – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
PO 001 – MAINTAIN MAIN BATTERY & ASSOCIATED SYSTEMS											
LG 001.01.01	450	Submarine Electrical Trainer/S-100 Classroom	UNC	1	6	1	0	0		1	-
EC 001.01	50	Submarine Electrical Trainer/S-100 Classroom	UNC	1	6	1	0	0		1	-
LG 001.02.01	100	Submarine Electrical Trainer/S-100 Classroom	UNC	1	6	1	0	0		1	-
EC 001.02	150	Submarine Electrical Trainer/S-100 Classroom	UNC	1	6	1	0	0		1	-
PO 002 – MAINTAIN DIESEL GENERATOR											
LG 002.01.01	UNK	Submarine Diesel Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
LG 002.01.02	UNK	Submarine Diesel	UNC	1	6	1	0	0		1	-

		Trainer / S100 Classroom									
EC 002.01	50	Submarine Diesel Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
LG 002.02.01	200	Submarine Diesel Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
EC 002.02	50	Submarine Diesel Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
PO 003 – MAINTAIN DC POWER DISTRIBUTION											
LG 003.01.01	350	Submarine Diesel Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
EC 003.01	50	Submarine Diesel Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
LG 003.02.01	100	Submarine Electrical Trainer /S-100 Classroom	UNC	1	6	1	0	0		1	-
EC 003.02	100	Submarine Electrical Trainer /S-100 Classroom	UNC	1	6	1	0	0		1	-

LG 003.03.01	100	Submarine Electrical Trainer /S-100 Classroom	UNC	1	6	1	0	0		1	-
EC 003.03	50	Submarine Electrical Trainer /S-100 Classroom	UNC	1	6	1	0	0		1	-
PO 004 – MAINTAIN MAIN PROPULSION SYSTEM											
LG 004.01.01	50	Submarine Electrical Trainer /S-100 Classroom	UNC	1	6	1	0	1		1	-
LG 004.02.01	UNK	Submarine Electrical Trainer /S-100 Classroom	UNC	1	6	1	0	1		1	-
LG 004.03.01	50	Propulsion Trainer/S-100 Classroom	UNC	1	6	1	0	1		1	-
EC 004.03	200	Propulsion Trainer/S-100 Classroom	UNC	1	6	1	0	1		1	-
PO 005 – MAINTAIN VC-AS											
LG 005.01.01	800	VC-AS Trainer/S100 Classroom	UNC	1	6	1	0	0		1	-
LG 005.01.02	800	VC-AS Trainer/S100 Classroom	UNC	1	6	1	0	0		1	-
EC 005.01	300	VC-AS Trainer/S100 Classroom	UNC	1	6	1	0	0		1	-

PC 005	300	VC-AS Trainer/S100 Classroom	UNC	1	6	1	0	0		1	-
PO 006 – MAINTAIN AC POWER GENERATION & DISTRIBUTION											
LG 006.01.01	250	S100 Classroom	UNC	1	6	1	0	0		1	-
EC 006.01	50	S100 Classroom	UNC	1	6	1	0	0		1	-
LG 006.02.01	250	S100 Classroom	UNC	1	6	1	0	0		1	-
EC 006.02	50	S100 Classroom	UNC	1	6	1	0	0		1	-
LG 006.03.01	50	S100 Classroom	UNC	1	6	1	0	0		1	-
PO 007 – MAINTAIN AUTO VOLTAGE & FREQUENCY REGULATORS											
LG 007.01.01	50	S100 Classroom	UNC	1	6	1	0	0			
PO 008 – MAINTAIN CSS & ALARM SYSTEMS											
LG 008.01.01	400	S-100 CSS Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
LG 008.02.01	250	S-100 CSS Trainer / S100 Classroom	UNC	1	6	1	0	0		1	-
LG 008.03.01	50	S-100 FDS Trainer / S100 Classroom/ Submarine Platform	UNC	1	6	1	0	0		1	-
LG 008.04.01	100	S100 Classroom/ Submarine Platform	UNC	1	6	1	0	0		1	-
PO 009 – CONDUCT LONG & SHORT PRE-SEAS											
LG 009.01.01	500	S100 Classroom/	UNC	1	6	1	0	0		1	-

		Submarine Platform									

ANNEX I – ON-THE-JOB PERFORMANCE REQUIREMENTS

Name and Initials **Rank** **SN**

Date Training Started **Expected Date of Completion** *(12 months from start date)*

This On-the-Job Performance Requirement (OJPR) is written IAW NPTORD 4500-55. The OJPR must be completed IAW the included instructions. Incorrect or incomplete OJPR submission will be returned for re-submission.

SUMMARY OF OJPR COMPLETION			
PC #	**Title**	**Date**	**TPO or TO** (Print & Sign)
001			PRINT NAME HERE SIGN NAME HERE
002			PRINT NAME HERE SIGN NAME HERE
003			PRINT NAME HERE SIGN NAME HERE
004			PRINT NAME HERE SIGN NAME HERE
005			PRINT NAME HERE SIGN NAME HERE
006			PRINT NAME HERE SIGN NAME HERE
007			PRINT NAME HERE SIGN NAME HERE
008			PRINT NAME HERE SIGN NAME HERE
009			PRINT NAME HERE SIGN NAME HERE
010			PRINT NAME HERE SIGN NAME HERE

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made for each trainee beginning On-the-Job Training (OJT). The master copy shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective (PO) to the required standard, the Training Petty Officer (TPO) or Training Officer (TO) will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. The Training Site (TS) unit CO is authorized to grant an initial extension at a maximum of 3 months. Subsequent extensions are granted IAW NPTORD 4500-55 and can only be granted by CNPTG.

4. If it is deemed not possible to conduct OJT for some POs (e.g., the necessary equipment is not fitted on a ship, required equipment is not on issue to the unit, or necessary equipment is otherwise unavailable), the TPO or TO are directed to contact a Standards representative (Training Development Officer) at the appropriate TE for guidance.

PROGRESS REVIEW

5. The TPO or TO is to regularly review the plan (minimum once every 30 days) to ensure that the trainee is progressing appropriately. The trainee shall be counselled on this review and any decisions that are made which will affect trainee progression.

6. Unsatisfactory progress is indicated by:

7. The trainee experiencing difficulty successfully completing the objectives; and/or

8. The trainee failing to complete this OJPR package within the allotted 12 month time period.

9. Trainees experiencing difficulty will be counselled on their performance.

10. Should a member require additional time to complete the OJPR, the TPO or TO will submit a request for approval on an extension of time IAW NPTORD 4500-55 to their Chain of Command that will include:

11. Justification for delay in completion;

12. Estimation of time of completion (initial extension is 3 months max); and

13. Training plan to meet the new time estimate.

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

Section Head or Head of Department (HOD) **Date**

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the USQ Electrical Maintainer VIC. The member is qualified to carry out the duties of electrical maintainer on a VIC Class Submarine and is AIMM qualified.

_____

Div Officer, School Cmdt, or TS unit CO **Date**

OJPR REMARKS

14. OJPR remarks shall be used for the following information only:

15. Unit in which the training and assessment were carried out:

16. Any unusual circumstances which precluded compliance with an objective:

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
PO 001 Maintain Main Battery & Associated Systems				
EO 001.01 Battery PM				
001.01.001	Coordinate and conduct All Round Readings			
001.01.002	Coordinate and conduct Agitation test			
001.01.003	Coordinate and conduct Battery top up			
001.01.004	Coordinate and conduct a Battery Scrub			
001.01.005	Coordinate and conduct TCC			
001.01.006	Secure induction system under normal circumstances			
001.01.007	Test and embark distilled water			
001.01.008	Conduct milliamp leak test IAW C-24-691-000/NY-Z01			
PO 002 Maintain Diesel Generator				
EO 002.01 Maintain Diesel System				
002.01.001	Inspect speed probe			
002.01.002	Replace speed probe			
PO 003 Maintain DC Power Distribution				
EO 003.02 Main Power Bus Readings				
003.02.001	Participate as team member for Main power bus readings			
003.02.002	Conduct DC earth tracing procedures IAW VCSOP's			
PO 004 Maintain Propulsion System				
EO 004.02 Main Motor Breakdown Readings				

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
004.02.001	Participate as a team member for MM breakdown readings			
PO 005 Maintain VC-AS				
EO 005.01 Operate VC-AS				
005.01.01	Operate control surfaces			
PO 006 Maintain AC Power and Distribution				
EO 006.02 AC Motors and Starters				
006.02.001	Demonstrate the procedure for adjusting hydraulic control valves (Sirco)			
EO 006.03 SFCs & TRUs				
006.03.001	Conduct load balancing of TRUs			
PO 007 – Maintain Auto Voltage & Frequency Regulators				
EO 007.01 AVFRs				
007.01.001	Demonstrate understanding of AVFRs by locating components involved in feedback loop			
PO 008 – Maintain CSS & Alarm Systems				
EO 008.03 - FDS				
008.03.001	Conduct function test of FDS system			
008.03.002	Conduct test on FDS field devices			
EO 008.04 – Maintain Control Panel				
008.04.001	Inhibit a channel on Diesel Control Panel card to facilitate troubleshooting			
PO 009 – Conduct Long & Short Pre-Seas				
EO 009.01 Long & Short Pre-Seas				

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
009.01.001	Coordinate and conduct pre-sailing checks on 24V DC Grade 1 & 2 IAW VCSOP			
009.01.002	Coordinate and conduct pre-sailing checks on VPDC IAW VCSOP			
009.01.003	Coordinate and conduct pre-sailing checks on propulsion switchboard IAW VCSOP			
009.01.004	Coordinate and conduct pre-sailing checks on Auto-pilot/ VC-AS IAW VCSOP			
009.01.005	Coordinate and conduct pre-sailing checks on 440V 60Hz AC System IAW VCSOP			
009.01.006	Coordinate and conduct pre-sailing checks on 115V 60Hz AC System IAW VCSOP			
009.01.007	Coordinate and conduct pre-sailing checks on 200/ 115V 400Hz IAW VCSOP			
009.01.008	Coordinate and conduct pre-sailing checks on alarm systems IAW VCSOP			
009.01.009	Coordinate and conduct pre-sailing checks on surveillance system IAW VCSOP			

Consolidation Walkthrough Guide

The walkthrough at the end of the AIMM OJPR will measure the candidate's understanding of the material studied and the systems they have learned. In order to pass the walkthrough, the candidate will have two attempts. If on the first attempt, they miss 3 or more questions, they will be given the opportunity to seek the correct answers and retry the next business day. If on the second attempt they continue to answer questions incorrectly, they will be advised to complete further study prior to attempting again. On a third attempt, if the candidate misses 3 questions on one system, then this will count as a failure of walkthrough and a TRB will be initiated. Walkthroughs will be set at a maximum of one (1) hour.

1. The walkthrough shall consist of 2 parts. Part 1 is a set of questions pertaining to each system to include PM, CM, and safety. Part 2 is based on a troubleshooting scenario. Candidates won't need to access references for these questions, however, they will need to be able cite where to find the proper procedure.
2. Conducted by RQ-PO2 Mar Tech 00379-03 with both AILS and AIMM qualifications.
3. This walkthrough is not comprehensive of what to ask the candidate. It provides the minimum standard to assess the viability of the candidate.
4. On the checklist if the candidate's answer is unsatisfactory list reasoning in remarks.
5. Walkthrough format:

Description	**Date**	**Remarks**
Part 1 - General Technical Knowledge Minimum one question on each system in each of the following categories:		
PM		
CM		
Safety		
Part 2 – Scenario Based Problem		
One troubleshooting scenario		

Part 1 - General Technical Knowledge

a. Main Battery & Associated Systems:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.
b. Diesel Generator:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.
c. DC Power Distribution:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.
d. Main Propulsion System:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.
e. VC-AS:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.

f. AC Power Generation & Distribution:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.
g. Auto Voltage & Frequency Regulators:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.
h. CSS & Alarm Systems:
   The candidate will be asked a minimum of 3 questions, one on PM, one on CM, and one on safety.

Part 2 - Scenario Based Problem
The candidate will be given a troubleshooting scenario and will be asked to explain how they would react. The scenario can be taken from any troubleshooting incident experienced by the Chief Electrician. Reactions to the scenario are to include:

a. Safety to plant and personnel
b. Troubleshooting method utilized (half split method, sequential method)
c. Estimate time to repair
d. Location of replacement part or component

INSTRUCTIONS GIVEN TO TRAINEE

This will be a consolidation walkthrough based on content from PO 001 - PO 004 and PO 006 – PO 009.

The walkthrough shall consist of 2 parts. Part 1 is a set of questions pertaining to each system to include PM, CM, and safety. Part 2 is based on a troubleshooting scenario.

To pass, you must complete all performance in accordance with the standard.

References will not be provided.

Pass / Fail.

Do you have any questions before you start?

CONSOLIDATION WALKTHROUGH NOTES
This checklist is not comprehensive of what to ask the candidate. It provides the minimum standard to assess the viability of the candidate.

Consolidation Walkthrough Checklist

PROTECTED B (when complete)

				SAT	UNSAT	Remarks			
Part 1.General Technical Knowledge									
System	**PM**	**S**	**U**	**CM**	**S**	**U**	**Safety**	**S**	**U**
Main Battery		☐	☐		☐	☐		☐	☐
Diesel Generator		☐	☐		☐	☐		☐	☐
DC Power Distribution		☐	☐		☐	☐		☐	☐
Main Propulsion		☐	☐		☐	☐		☐	☐
VC-AS		☐	☐		☐	☐		☐	☐
AC Power Generation & Distribution		☐	☐		☐	☐		☐	☐
AVFRs		☐	☐		☐	☐		☐	☐
CSS & Alarms		☐	☐		☐	☐		☐	☐
Part 2. Scenario Based Problem									
Troubleshooting Scenario				☐	☐				
Positive Feedback				**Improvement Feedback**					
Overall Assessment – Circle One									
Met Standard				**Did Not Meet Standard**					
Assessor I have discussed the results of this assessment with the candidate.				**Candidate** I have been debriefed on the result of this assessment.					
Assessor	Name:			Date:			Signature:		
Candidate	Name:			Date:			Signature:		

PROTECTED B (when complete)

OJPR PC 001 – Title

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Ship or Shore.

Arrangements and Procedures

2. PC 001 will be a practical assessment based on content from QSP Title PO 001.

3. The trainee must achieve all SATs to pass this assessment.

4. References may be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

5. PC 001 will be a practical assessment based on content from QSP Title PO 001.

6. During this practical assessment, you will be required to demonstrate your ability to PC Title.

7. You are allowed any job aides that would be available during normal operation of the equipment.

8. You must achieve all SATs to pass this assessment.

PART III – PERFORMANCE CHECK NOTES

9. Assessment will occur on Ship or Shore.

10. Trainees must perform… add further details about the performance being observed if required.

11. It is understood that PC tasks may be supervised by different delegated signing authorities depending on the availability of personnel. The TPO/TO and the trainee must sign PC checklist 001 upon completion of all tasks.

PROTECTED B (when complete)

OJPR PC 001 PO TITLE			
Item	**SAT**	**UNSAT**	**Signing Authority** (print name and sign)
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE
	☐	☐	PRINT NAME HERE SIGN NAME HERE

The trainee must achieve all SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: **(Circle One)**	**Met Standard**	**Did Not Meet Standard**
Training Petty Officer/Training Officer I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)